MSG: Robust Multimodal Remote Sensing Image Matching Using Side Window Gaussian Space

Chongyue Zheng, Shanshan Li, Chengyou Wang, *Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*—Robust and accurate image matching and registration are foundational tasks for numerous applications. However, current methods often fail when dealing with multimodal remote sensing images (MRSIs) that exhibit significant spatial geometric differences (SGDs) and nonlinear radiometric differences (NRDs). To address these challenges, this article proposes a novel MRSI matching (MRSIM) method: matching using side window Gaussian space (MSG). MSG leverages an intuitive concept that human visual perception relies heavily on salient features at image edges for precise matching. Specifically, the proposed method: 1) constructs a multiscale side window Gaussian filter scale space (MSGSS) that preserves edge information at different scales while blurring the image; 2) enhances the repeatability of keypoints by performing corner detection on edge maps; 3) increases descriptor robustness by using second-order gradients combined with steerable filtering; and 4) further utilizes a two-stage matching strategy within a constrained search space and designs a new distance, making full use of densely distributed edge keypoints. Quantitative and qualitative experiments conducted on five datasets spanning 957 image pairs across nine multimodal types demonstrate that MSG outperforms nine advanced algorithms (six feature-based methods: SIFT, OS-SIFT, RIFT, CoFSM, HOWP, and POS-GIFT; and three deep learning-based methods: SuperPoint + SuperGlue, LoFTR, and ReDFeat). The results indicate that MSG achieved a number of correct matches (NCMs) much higher than the compared algorithms, with the highest success rate (SR), lowest RMSE, and good time efficiency while achieving both scale and rotation invariance. Codes are available at https://github.com/ZCYla/MSG**

***Index Terms*—Feature descriptor, multimodal image matching, nonlinear radiometric differences (NRDs), scale and rotation invariance, side window filter.**

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I. Introduction

With the rapid advancement of multisensor information processing technology, satellites equipped with various sensors provide unprecedented frequency and diversity of Earth surface image data [1]. These data types, such as optical, synthetic aperture radar (SAR), light detection and ranging (LiDAR), and infrared images, are crucial for addressing global challenges like climate change, natural disaster assessment, and urban planning. Consequently, multimodal remote sensing image (MRSI) interpretation has become a research hotspot. Compared to single-modal observations, MRSI provides richer information, driving significant social progress [1], [2], [3].

MRSI matching (MRSIM) or registration technology bridges the gap between different modalities, allowing images acquired under various sensors, times, and imaging conditions to be accurately aligned. This process is essential for applications such as image fusion [4], [5], change detection [6], [7], [8], image stitching, and 3-D reconstruction [9]. As remote sensing (RS) applications evolve, improving image matching accuracy and efficiency has become increasingly important. However, the multimodal nature of these images introduces additional challenges.

MRSIM faces two major challenges: spatial geometric differences (SGDs) and nonlinear radiometric differences (NRDs) [1]. SGD primarily stems from variations in satellite attitude and sensor resolutions, causing translation, rotation, and scale discrepancies. NRD arises from differences in imaging mechanisms and external environmental conditions, resulting in significant intensity disparities between images. These differences increase the matching difficulty, especially when dealing with substantial differences, such as those between SAR and optical images.

Recent research has proposed various innovative multimodal image matching techniques, from handcrafted region-based [10], [11] and feature-based [12], [13], [14], [15] methods to the deep learning algorithms [16], [17], [18], [19], continuously enhancing matching accuracy and efficiency. Nevertheless, the existing techniques still face challenges, including inadequate resistance to SGD and NRD and high processing costs.

The region-based methods are designed to match image regions by calculating a similarity metric between blocks of target and reference images. These methods rely on prior geographical information to eliminate geometric transformations. As a result, they are disabled to handle scale and

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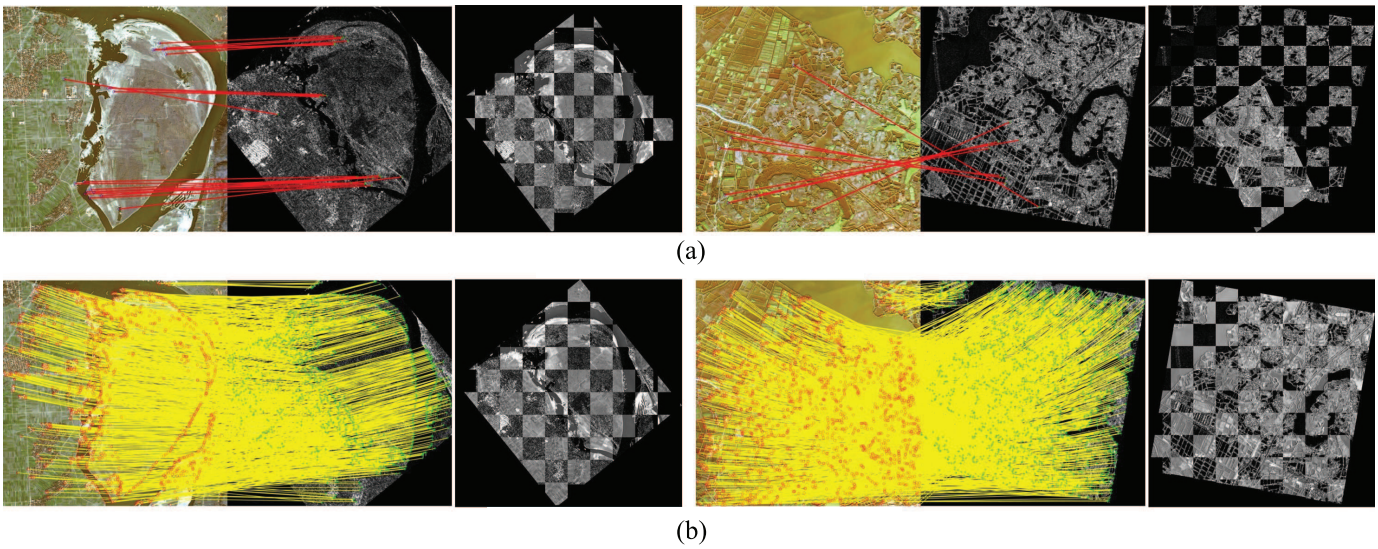

Fig. 1. Comparison between the feature-based method RIFT [26] and the proposed method MSG. These two examples (optical and SAR images) demonstrate that MSG effectively leverages the similar features on edge from different modals of RS images to guide the matching process. (a) RIFT matching results. (b) Proposed MSG matching results.

rotation differences, limiting their applicability in diverse conditions. The feature-based methods, such as SIFT [12], SURF [20], and ORB [21], try to extract scale and rotation-invariant features (SRIFs). Although these traditional image feature and primary orientation estimation methods are robust to linear intensity variations, they are sensitive to NRD and lack robustness when dealing with multimodal images with rotation differences. Consequently, they tend to produce more mismatches, leading to matching failures. Recent algorithms have addressed some of these issues by improving gradient features specific to RS images or by using phase congruency (PC) [22] to extract structural features. However, challenges remain in terms of computational efficiency and robustness under significant SGD and NRD conditions. The learning-based methods have shown promising results on relevant datasets, leveraging the powerful feature extraction and representation capabilities of deep learning models. However, their effectiveness is limited by the inherent complexities of MRSI and the lack of a comprehensive dataset that includes a sufficient number of labeled cross-modal RS images. Most learning-based studies and relevant datasets are tailored for the vision domain [23], [24], [25] and do not adequately address the unique SGD and NRD characteristics of RS images.

To address these limitations, this article proposes a novel feature-based MRSIM method: matching using side window Gaussian space (MSG). We observe that most feature-based matching methods focus on extracting more robust, modality-independent features from the original images, but they often fail to fully exploit the abundant edge information present in RS images to enhance matching accuracy. This oversight can limit performance. Given that different modalities often share similar structural features, which humans naturally prioritize during image matching, we were inspired to design the MSG method. As shown in Fig. 1, many keypoints along the edges, which are often missed during matching, are correctly matched in MSG. The algorithm utilizes the dense distribution of keypoints along the edges to optimize matching accuracy. These evenly distributed keypoints are then used to estimate the transformation matrix between the two images robustly, facilitating accurate registration and subsequent processes. The experimental results show that the proposed MSG method achieved higher matching accuracy and robustness under complex SGD and NRD conditions.

The key contributions of our proposed method are as follows.

1) The construction of a multiscale side window Gaussian filter scale space (MSGSS), which effectively preserves different levels of edge information while reducing noise and smoothing images using side window filter [27]. This approach allows for more reliable keypoint detection and feature extraction in multimodal images.
2) Keypoints are detected on binary edge maps, using adaptive nonmaximum suppression (ANMS) to ensure high repeatability and distinctive edge features. This significantly improves the probability of correct matches.
3) First-order Gaussian steerable filters are integration with the Sobel operator to compute second-order gradient magnitudes and phases, enhancing rotation invariance. This allows the construction of improved gradient location and orientation histogram (GLOH) descriptors, which are robust to NRD.
4) A two-stage matching strategy further increases robustness and accuracy. First, we apply the nearest neighbor distance ratio (NNDR) to obtain high-confidence matches. Next, a second matching is conducted using a newly designed distance metric within a constrained search space, refining the set of matching points.

The remainder of this article is organized into four sections. Related works are introduced in Section II. The methodology of the proposed MSG is described in Section III. Section IV designs experiments comparing MSG with other advanced

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methods, evaluating and analyzing the proposed algorithm's performance, and gives a detailed discussion of the experimental results. Finally, Section V summarizes the work with a conclusion.

II. Related Works

MRSIM is a complex task that integrates various technologies and methods. Typically, MRSIM algorithms can be classified into three categories: region-, feature-, and learning-based methods.

A. Region-Based Methods

Region-based (template-based) matching calculates the similarity between target and reference image blocks using predefined similarity metrics, selecting the most similar reference block as the correct match. Common similarity metrics include sum of squared differences (SSDs), normalized cross correlation (NCC), and mutual information (MI) [11]. Phase correlation methods [28] calculate frequency-domain similarities via Fourier transform, which are robust to illumination changes. However, these methods either suffer from low accuracy due to NRD or are time-consuming. To address this, the histogram of oriented PC (HOPC) [29] preserves image structure across different modalities but has limitations in accuracy and time efficiency. Channel features of oriented gradients (CFOGs) [10] generate pixel-level HOG-like orientation histograms using Gaussian convolution and use 3DFFT matching in the frequency domain, achieving rapid matching with excellent results. Steerable filters of first- and second-order channels (SFOCs) [30] utilize multiscale dilated convolution strategy and provide software for large-scale RS image registration. Despite some robustness to NRD, these methods are sensitive to scale and rotation differences and heavily rely on prior geometric information.

B. Feature-Based Methods

Feature-based methods typically include four stages: keypoint detection, descriptor extraction, feature matching, and outlier removal. Previous methods like SIFT [12], SURF [20], and ORB [21] primarily target linear intensity variations. However, multimodal image matching presents more difficulties. To overcome NRD limitations, researchers have proposed new algorithms. SAR-SIFT [31] uses multiscale ratio index weighted averaging to extract SAR image gradient features, resisting multiplicative noise interference. Position-scale-orientation SIFT (PSO-SIFT) [32] uses the second-order gradients and joint distance matching for enhanced NRD robustness. OS-SIFT [33] constructs two multiscale Harris scale spaces and limits gradient directions to address gradient reversal in multimodal images. Locally normalized image feature transform (LNIFT) [34] reduces image differences through local normalization. Co-occurrence filter space matching (CoFSM) [15] uses co-occurrence filter to construct a scale space and combines it with low-pass filtered to generate image gradients for optimized feature point extraction and descriptor construction. SRIF transform [35] defines a local intensity binary transformation to increase image similarity. Multiscale histogram of local main orientation (MS-HLMO) [36] uses mean squared gradients at different scales to obtain local orientation information. Modality-invariant feature transform with global-to-local searching (GLS-MIFT) [37] proposes adaptive partition descriptors and detects on multiscale, multidirectional filtered energy aggregation maps.

Additionally, PC [22] is often used to extract similar structural features across different modalities. The radiation-invariant feature transform (RIFT) [13], [26] utilizes logarithmic Gabor filtering to construct a max index map (MIM) to resist NRD. RIFT's end-to-end annular structure can handle angle rotational differences but struggles with scale differences. The 3MRS method [38] uses RIFT for coarse-to-fine two-step matching. The histogram of the orientation of the weighted phase (HOWP) descriptor [14] aggregates corner and blob feature points and establishes a new weighted phase orientation method to replace gradient direction features. The position-orientation-scale guided geometric and intensity-invariant feature transformation (POS-GIFT) [39] constructs Daisy descriptors for rotation-invariant to NRD using weighted norm information from multidirectional filter responses. It then reconstructs descriptors for matching based on initial computed transformation matrix parameters.

These methods primarily improve NRD robustness by defining various methods for extracting significant local features and their descriptor vectors from images. However, they continue to encounter challenges related to limited resistance to strong SGD and NRD, insufficient keypoint extraction, or high computational costs.

C. Learning-Based Methods

Deep learning significantly improves the accuracy and robustness of image matching tasks. Learning-based MRSIM methods can be divided into single-stage, full-stage, and style transfer methods.

Single-stage networks replace one matching stage with a deep learning module. D2-Net [40] uses local maxima mining as a detector and dense feature descriptors, though precision is poor due to detection on the feature map. SuperPoint [16] proposes a self-supervised framework pretrained on synthetic images, using a homography adaptation strategy for automatic annotation. SuperGlue [17] uses self-attention and cross-attention mechanisms in graph neural networks to learn complex relationships between descriptors. Meanwhile, some methods specifically target RS images. Map-Net [41] converts the matching task into image classification to extract high-level semantic information. ReDFeat [42] recouples feature detection and matching constraints with a large receptive field detector and a learnable nonmaximum suppression (NMS) layer. MU-Net [43] employs stacked deep neural networks to handle coarse-to-fine alignment across scales and uses a structural similarity-based loss to eliminate the reliance on labeled data. AESF [44] refines handcrafted structural features (e.g., CFOG) with a multibranch global attention module.

Full-stage networks utilize end-to-end architectures for image matching. LoFTR [18] adopts a detector-free dense matching algorithm using lightweight Transformer modules. MatchFormer [19] proposes a hierarchical extraction and

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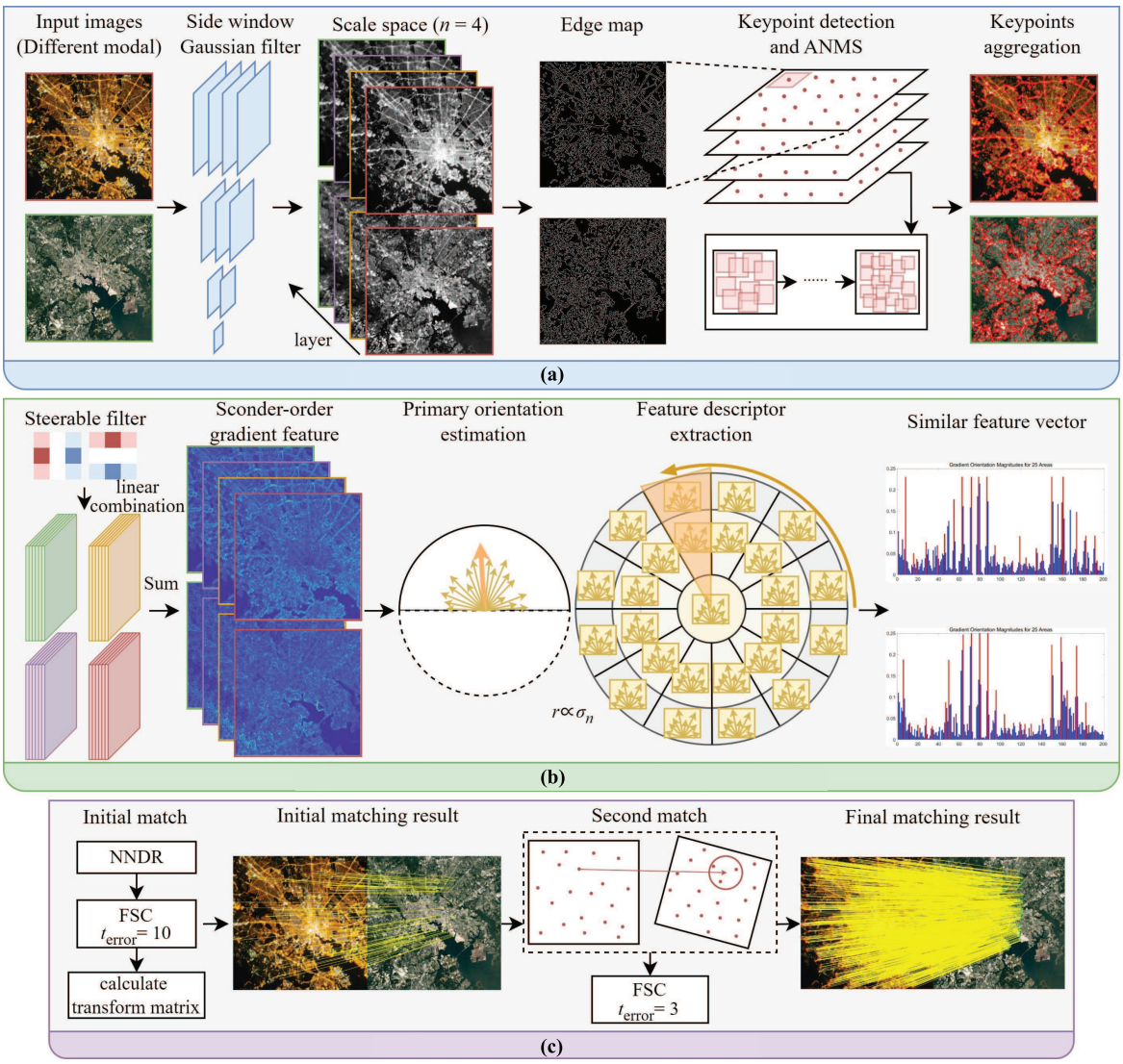

Fig. 2. Flowchart of the MSG. (a) MSGSS construction and keypoint detection. (b) Primary orientation estimation and descriptor extraction. (c) Feature match and mismatch removal.

matching framework, integrating self-attention and cross-attention of multiscale features. SemLA [45] embeds semantic information at each stage of the neural network for matching and fusion of infrared and visible images.

Style transfer methods address sample data limitations by converting different modal images into unified modal images. Generative adversarial networks (GANs) use an adversarial training mechanism to generate samples matching the real data distribution. Conditional GANs (CGANs) [46] combine with traditional handcrafted methods to generate SAR-like image blocks from optical images. *K*-means clustering guided (KCG)-GAN [47] uses *K*-means clustering analysis as input for image generation, improving synthetic optical image quality.

Despite their potential, current deep learning methods often rely on natural vision image datasets for training and testing [23], [24], [25], limiting their applicability to MRSIs. Furthermore, the lack of MRSIM datasets hinders their development and evaluation.

In summary, despite significant efforts, MRSIM still faces challenges due to NRD and SGD. Current methods, including region-, feature-, and learning-based approaches, have their strengths and limitations. Based on these observations, this article proposes a new feature-based method MSG, aiming to achieve high matching accuracy and robustness under complex conditions.

III. Methodology

The process flow of our MSG algorithm is illustrated in Fig. 2. The method comprises four key steps: construction of the MSGSS, keypoint detection, feature descriptor extraction, and two-stage feature matching with mismatch removal.

Initially, the MSGSS is constructed for the input images that have been preprocessed with Gaussian denoising, normalization, and contrast enhancement to generate a set of images with varying degrees of blurriness and edge features. Following this, Harris corner detection [48] is applied on

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the binary edge map extracted using the Sobel or Canny operator. The ANMS method suppression via square covering (SSC) [49] is then used to obtain highly repeatable keypoints with prominent edge information. Next, first-order Gaussian steerable filtering is applied in multiple directions, and the results are summed to obtain the first-order gradient map. The Sobel operator is used to compute the second-order gradient magnitude and phase. Based on this, the primary orientation is estimated, and improved GLOH descriptors are constructed for feature description. Finally, high-confidence matches are obtained using NNDR with a stringent threshold. The spatial constrained matching approach is then applied on the second stage, resulting in a refined set of matching points.

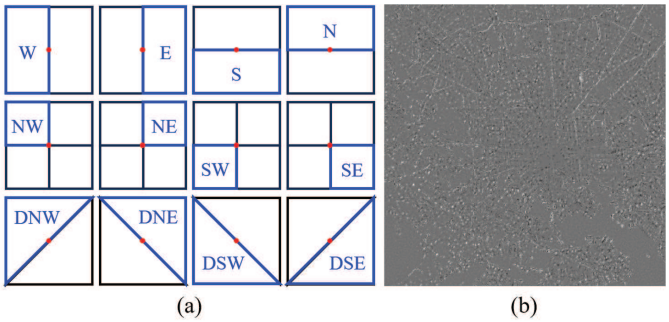

Fig. 3. (a) Filter kernel used by S-GAU. (b) Differences between 12-direction and 8-direction filtered images, $r = 3, \sigma = 1.6$, and iter = 1.

A. Multiscale Side Window Gaussian Filter Scale Space

In RS images, edge regions contain rich visual information. Effectively capturing edge features can significantly enhance matching performance, particularly under conditions of NRD. Traditional isotropic or anisotropic filters, while capable of smoothing and blurring images, often fail to simultaneously preserve edge features and remove noise, especially in NRD-affected MRSI. Therefore, constructing a new scale space that effectively preserves edge information and resists NRD is crucial. This motivated the development of the MSGSS.

1) Side Window Filtering Technology: Conventional local filters typically place the target pixel at the center of the window, estimating the output based on neighboring pixels. This often results in edge blurring, a fundamental issue for many filtering algorithms. Linear filtering of an image can be represented as

$$\hat{I}_i = \sum_{j \in \Omega_i} w_{ij} q_j \tag{1}$$

where $\hat{I}_i$ is the output intensity, q_j represents the intensity of neighboring pixels, and w_{ij} is the normalized weight determined by the filter kernel. The primary goal of filtering is to produce an output image close to the target image while maintaining similarity to the original image.

The side window filter technique [27] treats each target pixel as a potential edge and generates multiple local windows around it. Instead of placing the target pixel in the center, each window aligns the target pixel with a side or corner of the window. The output is a linear combination of neighboring pixels from the window with the smallest L_2 distance, effectively preserving edge characteristics while smoothing the image. Gaussian kernels and their derivatives are known to be the only feasible smoothing kernels in scale-space construction [50]. Based on this, the side window Gaussian filter (S-GAU) is developed, designed to blur the original image while preserving high-quality edge information. To further maximize the prominence of edge features in various directions, 12 specific windows are defined for discrete cases, as illustrated in Fig. 3. The 12-direction filtering produces clearer contours, and the discrepancies with the eight-direction filter results occur primarily at the edges of the image. The filter result of side window in direction $I'_{S-\mathrm{GAU}}$ is selected as output, which is given by

$$I'_{S-\mathrm{GAU}} = \underset{\theta \in \{W,E,\ldots,\mathrm{DSE}\}}{\arg\min} \left\| q_i - \sum_{j \in \Omega_i^\theta} w_{ij}^{\mathrm{GAU}} q_j \right\|_2^2 \tag{2}$$

where Ω_i^θ represents the local window centered on target pixel i, with Gaussian half kernels or quarter kernels applied in direction θ. w_{ij}^{GAU} denotes the Gaussian kernel weights normalized within each window, and q represents the intensity value on input image.

2) Scale Space Construction: In image matching tasks, images are typically upsampled or downsampled to generate the base layer of a pyramid. For instance, in the SIFT [12] scale space, the original image is downsampled between each octave, and Gaussian kernels with varying variances are applied between layers to achieve scale invariance. To reduce computational costs, this step was omitted in our approach. MSG uses descriptors with varying radii to achieve scale invariance. The radii and variance of the S-GAU filter are determined by the pyramid's level, with variance related to scale and proportional to feature descriptor sizes. MSG with one octave can handle scale differences of up to $2\times$. Increasing the number of octaves can handle larger scale differences, as shown in Section IV-D.

Gaussian filtering (GF) with the same radius and variance results in more blurred images than S-GAU. Therefore, MSGSS aims to create images with varying degrees of blur to simulate human visual perception. This is achieved by selecting different filter radii r and scale parameters θ_n for each layer to construct the scale space, adapting to various image features and details. The variation in capture features from coarse to fine scales, while the increase in filtering rounds iter refines the image at each scale, preserving key edge information. The parameters for different layers are chosen as follows:

$$\sigma_n = \sigma_0 \times 2^{n/(S-1)}, \quad (n = 0, 1, \ldots, S-1) \tag{3}$$

$$r = [3, 5, 8, 11], \quad \mathrm{iter} = [1, 2, 3, 4] \tag{4}$$

where $\sigma_0 = 1.6$ is the initial Gaussian kernel variance, S is the total scale layer number, and n is the layer index. The resulting images after Gaussian and S-GAU filtering are shown

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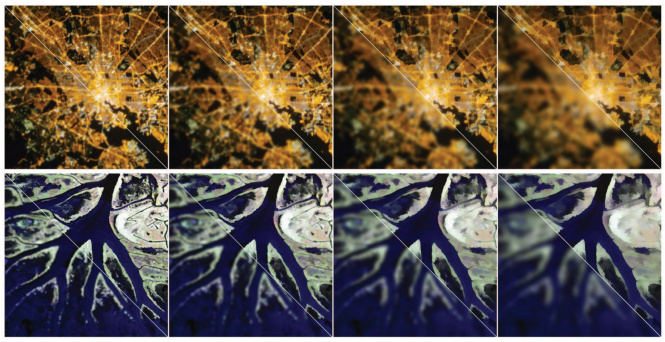

Fig. 4. Comparison of the results of GF (bottom left of each image) and S-GAU filtering (top right of each image) for nighttime optical images (first row) and SAR images (second row). The parameters of scale space are: $r = 3, 5, 8, 11$; $\sigma = 1.6, 1.6 \times 2^{1/3}, 1.6 \times 2^{2/3}, 3.2$; *and iter* $= 1, 2, 3, 4$.

in Fig. 4. It is evident that S-GAU preserves more edge details when constructing the scale space.

B. Keypoint Detection

Most multimodal matching methods [15], [33], [34] prioritize uniform distribution of keypoints, often neglecting the importance of prominent visual features. In this study, we enhance the probability of correct matching by extracting corner points with high repeatability and significant edge information from binary edge maps. Each filtered image in the scale space is processed using the Sobel or Canny operators to extract binary edges, followed by Harris corner detection [48] and ANMS. Finally, all detected keypoints are merged to increase the likelihood of successful matching. While the Canny operator [51] produces finer edges, detecting more points, the Sobel operator captures more prominent features. The choice between these methods depends on specific scenarios, with Sobel being the default method in this study.

1) Binary Edge Extraction and Harris Corner Detection: The Sobel operator is widely used to approximate the first derivative of an image, enhancing high-frequency components to identify boundaries. The Sobel filters for detecting edges in the x-direction ($\boldsymbol{G}_x$) and y-direction ($\boldsymbol{G}_y$) are defined as

$$\boldsymbol{G}_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \quad \boldsymbol{G}_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{5}$$

Convolution of an image with $\boldsymbol{G}_x$ and $\boldsymbol{G}_y$ produces the gradient images in the x- and y-directions, respectively. The gradient magnitude image $\boldsymbol{B}$ is computed as

$$\boldsymbol{B} = \sqrt{(\boldsymbol{I} * \boldsymbol{G}_x)^2 + \left(\boldsymbol{I} * \boldsymbol{G}_y\right)^2} \tag{6}$$

where $\boldsymbol{I}$ is the input image and * denotes convolution. A threshold t is set $t = (\alpha \cdot \text{mean}(\boldsymbol{B}))^{1/2}$, where pixels in $\boldsymbol{B}$ exceeding t are set to 1 and others to 0. NMS is then applied, resulting in a clearer binary edge image.

The Harris corner detection is known for its stability and efficiency, as it is less sensitive to intensity and scale variations. The Harris corner response R for each pixel is calculated as

$$R = \det(M) - \beta \cdot \text{trace}(M)^2 \tag{7}$$

where

$$M = \sum_{(u,v)\in W_\sigma} W_\sigma(u,v) \cdot \begin{bmatrix} I_x^2(u,v) & I_x(u,v) I_y(u,v) \\ I_x(u,v) I_y(u,v) & I_y^2(u,v) \end{bmatrix} \tag{8}$$

where $\det(\cdot)$ and $\text{trace}(\cdot)$ denote the determinant and trace of the matrix, respectively; β is an empirical constant; I_x and I_y are image gradients in the x- and y-directions; u and v are the x- and y-coordinates; and W_σ is a Gaussian window with variance σ. Pixels with response value R exceeding a threshold are considered corners.

2) Suppression via Square Covering: A significant challenge in MRSIM is the variability in image size and scale. During keypoint detection, it is crucial to use NMS to filter out clustered, redundant, and noisy points while retaining the most relevant keypoints. This study employs the SSC [49] method, which achieves fast and uniform keypoint distribution by covering keypoints with square regions. The key steps of SSC are listed as follows.

1) *Keypoints Sorting:* Keypoints set $\boldsymbol{P}_{\text{in}}$ is sorted based on their intensity values obtained in the keypoint detection procedure.

2) *Initialization of Binary Search Boundaries:* To accelerate convergence, upper and lower boundaries a_{h} and a_{l} for the binary search are calculated based on image dimensions, input points, desired output points, and tolerance

$$a_{\text{h}} = \frac{-H_{\text{I}} + W_{\text{I}} + 2m - \sqrt{\Delta}}{2(m-1)} \tag{9}$$

$$a_{\text{l}} = \frac{1}{2}\sqrt{\frac{n}{m}} \tag{10}$$

where $\Delta = 4W_{\text{I}} + 4m + 4H_{\text{I}}m + H_{\text{I}}^2 + W_{\text{I}}^2 - 2W_{\text{I}}H_{\text{I}} + 4W_{\text{I}}H_{\text{I}}m$; W_{I} and W_{I} are the image width and height, respectively; m is the desired number of output points; and n is the total number of detected points.

3) *Binary Search and Suppression:* In each binary search iteration, the grid resolution G_ω is set to $\omega/2$, where is the current suppression length. Each keypoint p_i is checked to see if it falls within a covered grid cell G_ω. Uncovered keypoints are added to the output set $\boldsymbol{P}_{\text{out}}$ and marked as covered within a 2ω length square region centered on p_i.

4) *Output Result:* The process iterates, adjusting until the desired number of keypoints or a termination condition is met, yielding a uniformly distributed keypoints set $\boldsymbol{P}_{\text{out}}$.

The SSC selects keypoints without calculating Euclidean distances between them. This optimization is particularly vital for matching large-scale RS images in a real application scene. Dynamic adjustment of the search range and iterative process ensures the quality and uniformity of keypoints, providing a robust set of feature points for subsequent matching.

C. Feature Calculation and GLOH Descriptor Construction

In MRSIM, NRD often leads to inconsistent gradient orientations and magnitudes at the corresponding points in different images. Typically, feature vectors are computed based on gradient orientations and magnitudes distributed from 0° to

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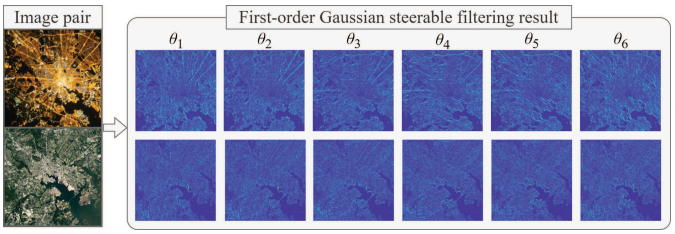

Fig. 5. Multidirectional first-order Gaussian steerable filtering results.

360° around keypoints. This tends to make achieving close matches between feature descriptors unreliable. To enhance descriptor's robustness against NRD, MSG uses the second-order gradients obtained through steerable filtering to assign primary orientations to keypoints and construct an improved GLOH descriptor.

1) Combined Steerable Filtering and Second-Order Gradient: To capture consistent structural information and local curvature attributes across different modalities, we employ a combination of steerable filters [52] that are linearly combined, and their absolute values are summed to obtain the first-order gradient image. This is followed by using the Sobel operator to calculate second-order gradients.

Steerable filters allow the generation of any directional filter through linear combinations of base filters, enabling the detection of specific directional patterns and capturing more directional and edge information. The Gaussian function and its higher order derivatives are known to be steerable [48]. The Gaussian function $G(x, y)$ with variance σ and its first-order Gaussian steerable base filter kernels $G^x_{1,\sigma}$ and $G^y_{1,\sigma}$ in the 0° and 90° directions are expressed as

$$\begin{cases} G(x,y) = \dfrac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \\ G^1_x(x,y) = \dfrac{\partial G}{\partial x} = \dfrac{-x}{2\pi\sigma^4} e^{-\frac{x^2+y^2}{2\sigma^2}} \\ G^1_y(x,y) = \dfrac{\partial G}{\partial y} = \dfrac{-y}{2\pi\sigma^4} e^{-\frac{x^2+y^2}{2\sigma^2}}. \end{cases} \tag{11}$$

Filtering an image $I(x, y)$ with the first-order base filters yields $R^x_{1,\sigma}$ and $R^y_{1,\sigma}$. By linearly combining these results, the filtered image $R^{\theta_j}_{\sigma_i}$ at scale σ_i and direction θ_j is obtained

$$\begin{cases} R^x_{1,\sigma} = G^x_{1,\sigma} * I(x,y), R^y_{1,\sigma} = G^y_{1,\sigma} * I(x,y) \\ R^{x,\theta}_{\sigma}(x,y) = \cos(\theta_j) R^{x,\sigma}_{\sigma}(x,y) + \sin(\theta_j) R^{y,\sigma}_{\sigma}(x,y) \\ \sigma_i = \tau \cdot i, \quad i \in [1, N_\sigma] \\ \theta_j = \dfrac{\pi(j-1)}{N_\theta} \end{cases} \tag{12}$$

where N_σ denotes the total number of scale layer, N_θ represents the number of filter directions, and τ is the empirical parameter. In this study, N_θ is set to 6 and τ to 0.8. Directional information is crucial for rotation invariance in image matching. Fig. 5 shows the filtering results in six directions. By summing the absolute value of filtering results from different directions, the first-order multidirectional steerable filtering

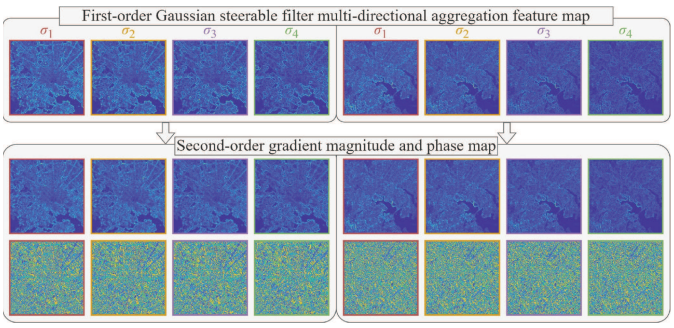

Fig. 6. Second-order gradient magnitude and phase map obtained by multiscale steerable filter aggregation.

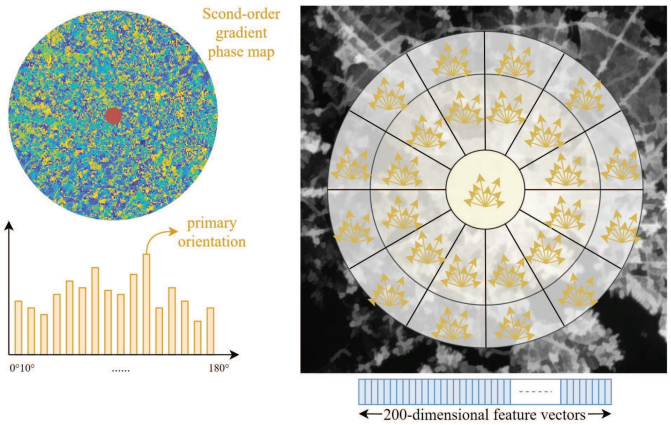

Fig. 7. Primary orientation estimation and improved GLOH descriptor construction.

results $F_\sigma(x, y)$ for each scale are obtained, which can be represented as the following equation:

$$F_{\sigma_i}(x,y) = \sum_{j=1}^{N_\theta} \left| R^{\theta_j}_{\sigma_i}(x,y) \right|. \tag{13}$$

Subsequently, the Sobel operator is used on $F_{\sigma_i}(x, y)$ to obtain second-order gradient magnitude $G_{2,\sigma}$ and phase images $\phi_{2,\sigma}$, which are then used in the descriptor construction. Considering the common occurrence of gradient reversal in multimodal images [32], the phase is unified to $[0°, 180°)$. Specifically, the values exceeding 180° will be subtracted by 180°. The final results are illustrated in Fig. 6.

2) Primary Orientation Estimation and Improved GLOH Descriptor: This process is illustrated in Fig. 7. During the primary orientation estimation step, $\phi_{2,\sigma}$ are averaged into 18 orientation intervals based on the phase values. GF with variance $2\sigma_n$ is applied to $G_{2,\sigma}$ to emphasize gradient orientation information closest to the current keypoint. Then, the gradient orientation histogram feature is collected within the keypoint's $6\sigma_n$ scope for the current scale. The orientation corresponding to the maximum value is designated as the primary orientation. Finer orientations can be obtained by interpolating with the two neighboring orientations. The descriptor is then rotated to align with the primary orientation, and feature vectors will be extracted starting from the region where the descriptor was constructed in that orientation, eliminating rotation differences.

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Previous studies have demonstrated the excellent performance of GLOH descriptors in MRSIM [32]. In this study, we use log-polar sectors similar to GLOH to create feature descriptors. Each keypoint is centered within a circular region with a radius of $R_s \times \sigma_n$ pixels, defining the positive rotational direction based on the image coordinate system, where R_s is a parameter that can be adjusted. Descriptors of different radii are constructed to achieve scale invariance. Each circular ring region is divided into N_0 tangential sectors, each representing a portion of the gradient orientation histogram within the feature neighborhood. Local gradient orientation values in each sector are quantized into N_{bin} bins, evenly distributed within $[0°, 180°)$. The final descriptor is a $(2 \times N_0 + 1) \times N_{\text{bin}} = 200$ dimensional vector, providing a detailed representation of local image structures. The specific parameter settings for the descriptor are discussed in Section IV-B.

D. Matching and Mismatching Removal

In this study, we propose a two-stage matching method to enhance the reliability of matched points. By combining NNDR for initial reliable matches, a new matching distance and a refined search space approach, the proposed method effectively reduces mismatches and missed matches, increase the number of matched points, and further improve registration accuracy.

Let $\boldsymbol{S}$ and $\boldsymbol{R}$ be the feature sets of two different modal RS images. The matching problem can be formulated as finding a mapping $M : F_1 \rightarrow F_2$ that minimizes the matching error E

$$E(M) = \sum_{f_1 \in F_1} \|\phi(f_1) - \phi(M(f_1))\|_2^2 \tag{14}$$

where $\phi(\cdot)$ represents a feature descriptor and $M(f_1)$ is the f_1 feature's corresponding matching feature in F_2.

Given two sets of keypoints extracted from the images to be matched, denoted as $\boldsymbol{S}$ and $\boldsymbol{R}$, the matching process is conducted in two stages. In the first stage, we apply the NNDR with a strict threshold 0.95 to identify the most similar matching points based on descriptor distances measured by cosine similarity. This ensures that the matched keypoints in $\boldsymbol{S}$ and $\boldsymbol{R}$ are highly reliable. The initially matched points form a preliminary correspondence set $\boldsymbol{C}_h$. This set $\boldsymbol{C}_h$ is then input into the fast sample consensus (FSC) [53] model to remove outliers, with the error threshold t_{error} set to 10 pixels to estimate the transformation model parameters.

In the second stage, we focus on refining the initial transformation model and conducting a more precise search for each keypoint within its surrounding space. Each keypoint in $\boldsymbol{S}$ is mapped to its corresponding location in $\boldsymbol{R}$, and the nearest neighbors within a specified search radius are considered for matching. This results in a refined correspondence set $\boldsymbol{C}_s$. In this study, for each point in $\boldsymbol{S}$ transformed to $\boldsymbol{R}$, the three nearest points in $\boldsymbol{R}$ are selected as the candidate matches. The most similar point is then chosen. Considering that the displacement direction of adjacent keypoints should be roughly similar [15], we redefine the matching distance D_{new} in the second stage as a combination of positional distance and descriptor feature distance

$$D_{\text{new}}\left(P_1^i, P_2^j\right) = \text{cosSim}\left(D_1^i, D_2^j\right) \cdot \left(1 + \left\|P_1^i - T\left(P_2^j, \mu\right)\right\|_2\right) \tag{15}$$

where cosSim$(\cdot)$ is the cosine similarity operation, P_1 and P_2 represent the matching points' position in two images, D_1 and D_1 are the feature vectors, T denotes the transformation model, and μ represents the model parameters obtained in initial matching. The refined set $\boldsymbol{C}_s$ is input into FSC again for further outlier removal, with t_{error} set to 3 pixels to obtain more accurate matching points.

TABLE I
NUMBER OF MATCHED POINTS AT DIFFIRENT FEATURE MATCHING STAGES ON DATASET 1 RS TYPES

Matching stage	Matched points number
After the first match	965
After the first FSC	217
After the second match	5365
After the second FSC	1003

Table I shows that after the first round of outlier removal, a large number of detected keypoints remain unmatched. The second stage leverages the results of the first stage to match as many keypoints as possible between the two images. This significantly improves the matching accuracy and success rate (SR).

IV. EXPERIMENTAL RESULTS

To evaluate the performance of the proposed MSG, we conducted four groups of experiments.

1) *Parameter Sensitivity Analysis:* This experiment tested the sensitivity of MSG to different parameters to determine the optimal settings.
2) *Quantitative and Qualitative Comparisons:* The performance of MSG was compared with eight other methods on five datasets, including both feature-based methods (SIFT [12], OS-SIFT [33], RIFT [26], CoFSM [15], HOWP [14], and POS-GIFT [39]) and deep learning-based methods (SuperPoint [16] + SuperGlue [17], LoFTR [18], and ReDFeat [42]). The specific settings for these algorithms are provided in Table II. Note that all methods were used with the default settings and pretrained models provided by the authors with the affine transformation model applied for outlier removal. For deep learning methods, we used FSC [53] with affine transformation model and an error threshold of 3 pixels on the initial matching points to obtain the output matching points. The time efficiency of all feature-based methods was also evaluated.
3) *Rotation and Scale Invariance:* This experiment demonstrated the rotation and scale invariance of the proposed method on image pairs with varying degrees of rotation and scale differences. The effect of different scale-space strategies on performance was also tested.

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TABLE II
SPECIFIC SETUP OF THE MSG AND NINE ALGORITHMS COMPARED IN THIS ARTICLE

Method	Main parameters	Implementations
SIFT	Keypoint number: 5000; Patch size: $15\sqrt{2} \times$ scale, Grids number: 4×4, Orientation bins: 8.	https://www.vlfeat.org/overview/sift.html
OS-SIFT	Keypoint number: 5000; Patch size: $24 \times$ scale, Grids number: 19, Orientation bins: 8.	https://github.com/xym2009/OS-SIFT
RIFT	Keypoint number: 5000; Patch size: 96, Grids number: 6×6, Orientation bins: 6.	https://github.com/LJY-RS/RIFT-multimodal-image-matching
CoFSM	Co-occurrence filter window size: 5; Scale layers: 4; Grids number: 19, Orientation bins: 8.	https://github.com/yyxgiser/CoFSM
HOWP	NMS window size: 3; Log-Gabor filter's scale number: 3, Orientation bins: 6; Descriptor size: 38.	https://skyearth.org/publication/project/HOWP/
POS-GIFT	Log-Gabor filter's scale number: 4, Orientation bins: 6; Concentric circles number: 3, Number of directions per circle: 12, Radius of first circle: 6; Number of nearest neighbors in POS strategy: 20.	https://github.com/Zhuolu-Hou/POS-GIFT
SuperGlue	Feature Extractor: SuperPoint; Keypoint number: 2048; Scene: "outdoor"; Image size: max size resampled to 640.	https://github.com/magicleap/SuperGluePretrainedNetwork
LoFTR	Backbone: ResNetFPN; Scene: "outdoor"; Image size: resampled to be divisible by 8.	https://github.com/zju3dv/LoFTR
ReDFeat	Model: Visble-SAR; Keypoint number: 4096; Scale factor: $2^{1/4}$.	https://github.com/ACuOoOoO/ReDFeat
MSG (ours)	Side window filter radius: $r = [3, 5, 8, 11]$; Iteration $= [1, 2, 3, 4]$; Max keypoint number per lay: 2000; Patch size: $12/20 \times$ scale, Grids number: 25, Orientation bins: 8.	https://github.com/ZCYla/MSG

4) *Ablation Study:* We conducted ablation experiments, replacing steps in MSG with different strategies for comparison to explore the effectiveness of each component in MSG.

For quantitative evaluation, we use four metrics: SR, number of correct matches (NCMs), root-mean-square error (RMSE), and mean time (MT). SR reflects the robustness of the matching method; NCM indicates the strength of the matching capability; RMSE reflects the matching accuracy, with lower values indicating higher accuracy; and MT measures the algorithm's time efficiency. Additionally, matching points with a real matching pixel error (E) greater than 5 pixels were considered mismatched, and image pairs with NCM less than 4 or RMSE greater than 6 were considered matching failures, with the RMSE value for failed matches set to 6. The formulas for SR (r_s), E, and RMSE (e_{rms}) are as follows:

$$r_s = \frac{N_{success}}{N_{total}} \times 100\% \tag{16}$$

$$E = \sqrt{\left[T(x_1, y_1) - (x_2, y_2)\right]^2} \tag{17}$$

$$e_{rms} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[H\left(x_1^i, y_1^i\right) - \left(x_2^i, y_2^i\right)\right]^2} \tag{18}$$

where $N_{success}$ and N_{total} represent the number of successfully matched image pairs and the total number of image pairs used for matching, respectively. For each image, more than ten evenly distributed corresponding points were manually annotated, which are considered as the ground truth, represented as (x_1^i, y_1^i) and (x_2^i, y_2^i). (x_1, y_1) and (x_2, y_2) represent the matching points' coordinates obtained by the algorithm. N denotes the number of annotated corresponding points. T denotes the transformation performed by the ground truth matrix calculated from manually annotated corresponding points. H denotes the transformation performed by matrix calculated from the output matching points.

The experiments were conducted using MATLAB R2019a on the following platform: Intel Core i7-10750H (six cores), running at a frequency of 2.60 GHz, with 16 GB of RAM, on a Windows 10 $\times$ 64 operating system.

A. Dataset Introduction

To evaluate the performance and robustness of the proposed MSG method, we compiled five distinct multimodal image datasets, totaling 957 image pairs. These datasets cover diverse real-world RS scenarios and associated challenges. Fig. 8 shows the representative image pairs from each dataset. The detailed information for each dataset is provided as follows.

1) Dataset 1: It contains 107 diverse multimodal image pairs from both RS and computer vision (CV) domains. It sourced from multiple literature studies [3], [29], [54]. The RS portion (66 pairs) includes seven modality combinations: Optical-DSM, Optical-Infrared, Optical-Map (10 each), Optical-SAR, Day-Night (11 each), Optical-Depth, and Multitemporal Optical (7 each). These cover extensive practical RS scenarios, presenting significant challenges such as severe NRD, SGD, and temporal changes. The CV portion (RGB-NIR [3]: 30 pairs and VIS-IR [55]: 11 pairs) comprises indoor/outdoor scenes, highlighting cross-modality matching challenges under varied illumination.

2) Dataset 2: It consists of 300 high-resolution (1-m) Optical (Google Earth) and SAR (Gaofen-3) urban image pairs (512 $\times$ 512 pixels) [56]. The primary challenges are severe NRD and complex SGD (including layover/perspective distortion) due to high-resolution SAR's speckle noise and scattering properties in urban areas. High-density textures and

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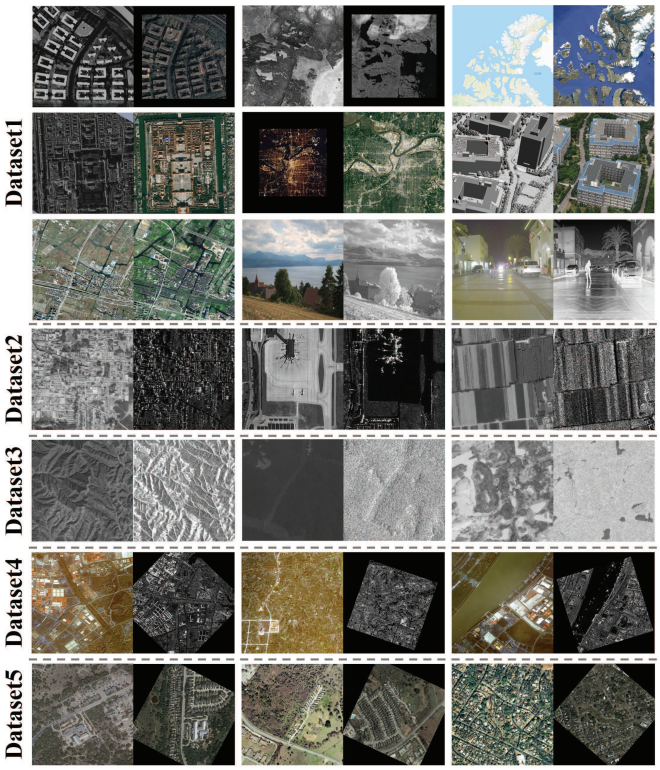

Fig. 8. Some multimodal image pairs in each dataset.

simple regions coexist, demanding high accuracy for practical urban registration.

3) Dataset 3: It includes 200 pairs of Sentinel-1 SAR and Sentinel-2 Optical images with medium-to-low resolution (10 m) [57]. It derived from the SEN1-2 dataset. Common in environmental monitoring and land cover mapping, this dataset challenges arise from Sentinel-1/2 radiometric differences, limited detail at lower resolution requiring robust structural feature utilization, and potential temporal changes between acquisitions. It offers global coverage across varied terrains (forests, deserts, and lakes).

4) Dataset 4: It sourced from the WHU-OPT-SAR dataset [58] provides 150 pairs of Optical (GF1 2 m)-SAR (GF3 5 m) images, cropped to 1024×1024 and resampled to 512×512. Used for image classification, this dataset presents typical Optical-SAR matching challenges across different terrain and vegetation types.

5) Dataset 5: It contains 200 pairs of multitemporal optical images [59] (512×512 after resampling). Primarily used for change detection, its main challenges stem from radiometric differences due to temporal changes (e.g., vegetation cycles and building alterations) and varying illumination/shadows.

To increase the diversity of the experimental data, random transformations (translations: −30 to 30 pixels, scaling: 0.7–1, rotations: 0°–90°) were applied for Datasets 4 and 5. The ground truth transformation matrices and sampled points were recalculated accordingly. For all images, 25 points uniformly distributed across each image were chosen to estimate RMSE.

TABLE III

PERFORMANCE OF MSG ON DATASET 1 RS TYPES WITH DIFFERENT PARAMETER SETTINGS (BOLDING INDICATES OPTIMAL)

Tested Parameters	Parameter Setting				r_s (%)↑	N_{cm}↑	e_{rms}↓	t_m (s)↓
	N_{bin}	N_0	R_s	K				
N_{bin}	6	12	12	2000	98.5	**920**	1.76	9.48
N_{bin}	8	12	12	2000	**100.0**	907	**1.69**	**9.01**
N_{bin}	10	12	12	2000	95.5	886	2.00	9.87
N_{bin}	12	12	12	2000	95.5	881	2.03	10.39
N_0	8	8	12	2000	97.0	901	1.92	9.23
N_0	8	10	12	2000	97.0	900	1.89	9.06
N_0	8	12	12	2000	**100.0**	907	**1.69**	**9.01**
N_0	8	14	12	2000	97.0	907	1.87	9.71
N_0	8	16	12	2000	97.0	**908**	1.83	9.87
R_s	8	12	8	2000	84.8	818	2.39	9.16
R_s	8	12	12	2000	**100.0**	**907**	**1.69**	**9.01**
R_s	8	12	16	2000	**100.0**	898	1.85	9.88
R_s	8	12	20	2000	**100.0**	863	1.86	12.20
K	8	12	12	1000	**100.0**	628	1.81	**6.75**
K	8	12	12	1250	98.5	747	1.84	7.64
K	8	12	12	1500	**100.0**	829	1.73	8.06
K	8	12	12	1750	**100.0**	888	**1.69**	8.61
K	8	12	12	2000	**100.0**	907	**1.69**	9.01
K	8	12	12	2250	**100.0**	924	**1.69**	9.61
K	8	12	12	2500	**100.0**	**933**	**1.69**	9.17

B. Parameter Settings

To comprehensively evaluate the performance of MSG, we analyzed four core parameters: the number of gradient directions for the descriptor (N_{bin}), the number of tangential partitions (N_0), the descriptor radius parameter (R_s), and the max number of keypoints per layer (K). These parameters were evaluated using four metrics: SR, NCM, RMSE, and MT. Since Dataset 1 covers most of the RS data types and has varying degrees of geometric transformations, the experiments in this section are performed on Dataset 1 to determine the most suitable parameters.

N_{bin} and N_0 determine the number of gradient directions used in constructing the feature vector and the number of partitions along the tangential direction within the descriptor. Higher values provide more detailed directional information. R_s influences the size of the region around each keypoint used to construct the descriptor. K sets the upper limit for the number of keypoints extracted per layer. Higher values improve robustness but increase computational load.

The specific experimental settings are shown in Table III. When N_{bin} is set to 8, both SR and NCM reached their maximum values, and RMSE was minimized, while MT remained at a low level. As N_0 increased from 8 to 12, RMSE gradually decreased, and MT slightly declined. However, when N_0 exceeded 12, both MT and RMSE increased, and only with N_0 did MSG achieve successful matching across all images. Failures occurred in two image pairs due to strong NRD with sparse texture in Optical-Map images. When R_s was set to 8, many images failed to match due to small descriptor sizes. Increasing R_s to 16 or 20 led to higher RMSE and MT, while R_s minimized RMSE and maximized NCM. Additionally, as K increased from 1000 to 2500, NCM gradually rose, and RMSE decreased, but MT also increased. When K exceeded 1750, RMSE no longer changed, and NCM slightly increased, while MT also slightly increased. This suggests that the number of keypoints extracted from binary edge maps using the Sobel

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TABLE IV

QUANTITATIVE COMPARISON OF EXPERIMENTAL RESULTS ON DATASETS 1–4 (BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST PERFORMANCE AMONG ALL METHODS, RESPECTIVELY. GRAY HIGHLIGHTS INDICATE THE BEST RESULT WITHIN THE RESPECTIVE GROUPS OF COMPARED METHODS AND OUR METHOD)

Dataset ID	Metric	Tested method										
		SIFT	OS-SIFT	RIFT	CoFSM	HOWP	POS-GIFT	SuperGlue	LoFTR	ReDFeat	MSG (ours)	MSG* (ours)
Dataset 1 (RS)	$r_s(\%)$	37.9	56.1	63.6	74.2	95.5	97.0	97.0	84.8	92.4	**100.0**	98.5
	N_{cm}	11	29	77	221	132	1275	142	298	138	907	**1492**
	e_{rms}	4.64	4.21	3.63	3.47	2.28	2.27	1.93	2.81	2.18	**1.69**	2.37
Dataset 1 (CV)	$r_s(\%)$	73.2	92.68	75.6	61.0	97.6	85.4	97.6	92.7	95.1	**100.0**	97.6
	N_{cm}	284	519	498	212	551	1664	238	**3316**	382	1686	1584
	e_{rms}	2.87	2.31	3.03	4.29	2.44	2.02	2.00	2.19	2.20	**1.85**	2.25
Dataset 2	$r_s(\%)$	0.0	4.7	26.3	5.3	50	29.7	26.3	6.0	79.3	79.3	**96.3**
	N_{cm}	0	3	29	30	36	291	10	2	83	910	**1947**
	e_{rms}	6.00	5.92	5.50	5.90	4.95	5.38	5.32	5.75	3.53	3.71	**3.01**
Dataset 3	$r_s(\%)$	0.0	16.5	17.5	11.0	41.5	45.0	10.5	37.5	63.0	85.5	**92.5**
	N_{cm}	0	5	18	55	20	429	6	15	25	965	**1959**
	e_{rms}	6.00	5.69	5.57	5.67	4.84	4.81	5.58	4.37	3.81	**2.67**	2.75
Dataset 4	$r_s(\%)$	0.0	1.3	9.3	5.3	9.3	49.3	20.0	27.3	26.7	85.3	**88.7**
	N_{cm}	0	1	11	14	4	536	25	21	32	1169	**2459**
	e_{rms}	6.00	5.99	5.80	5.94	5.75	4.47	5.07	4.88	4.82	2.79	**2.59**
Dataset 5	$r_s(\%)$	34.0	51.5	30.5	36.5	26.5	68.0	65.0	50.5	25.0	**84.5**	83.0
	N_{cm}	13	21	57	109	28	961	134	297	40	1034	**1999**
	e_{rms}	4.80	4.41	5.12	5.07	5.09	3.63	3.24	3.79	4.99	**2.54**	3.01

operator generally does not exceed 1750 per layer. Considering varying image sizes in practical applications, setting K to 2000 ensures that more points are extracted in large or texture-rich images, increasing NCM.

Based on the above analysis, the default parameters for MSG are set to $N_{\text{bin}} = 8, N_0 = 12, R_s = 12$, and $K = 2000$ to ensure optimal accuracy and efficiency. Additionally, R_s is fine-tuned based on the characteristics of each dataset. For Dataset 1, increasing the descriptor radius slightly raises RMSE and consumes more MT. Therefore, we retained the above radius to balance accuracy and computational efficiency. Because Datasets 2 and 3 feature more speckle noise and complex scenes with areas of low texture and structure, and Datasets 4 and 5 involve more geometric transformations, we increased R_s to 20 for Datasets 2–5 to enhance the model's robustness. Moreover, in practical applications, whether to use the Canny operator for extracting more keypoints can be selected depending on the specific scenario.

C. Quantitative and Qualitative Evaluation

1) Quantitative Evaluation: To demonstrate the performance of each method, we compared the quantitative results on all datasets. Based on the results, MSG achieved the best performance. The average experimental results on five datasets are presented in Table IV. MSG* denotes the results with Canny edge extraction and $R_s = 30$. For Dataset 1, MSG was the only method to achieve successful matching across all images, with the lowest RMSE within 2 pixels and NCM significantly higher than most of the other methods. On Datasets 2–5, MSG outperformed almost all other methods across all metrics, which clearly demonstrates the superiority of the method proposed in this article. Detailed comparisons of specific metrics across nine types of Dataset 1 are provided in Table V. Overall, the quantitative results are consistent with the qualitative results discussed afterward. Note that for the VIS-IR dataset, the Canny operator was used for edge detection.

As shown in Table IV, the image quality in Dataset 1 is relatively higher than that in Datasets 2–5. This enables deep learning-based methods to outperform several traditional feature-based methods, including SIFT, OS-SIFT, RIFT, and CoFSM. Dataset 2 exhibits a greater challenge due to its high resolution. Dataset 3 features a broader range of land cover types and weaker textures, thereby limiting the discernible feature information within each image. Datasets 4 and 5, employed for change detection, incorporate temporal variations alongside geometric transformations. SIFT failed to produce matches on Datasets 2–4, attributable to its difficulties with multimodal data. OS-SIFT succeeded on only a limited number of images. HOWP achieved the third-highest SR on Dataset 2, trailing MSG and ReDFeat. However, its performance, along with RIFT, deteriorated significantly on Datasets 4 and 5 due to the geometric transformations involved. CoFSM demonstrated subpar performance on the more challenging Optical-SAR image pairs. In contrast, POS-GIFT yielded comparatively better results than other feature-based methods, securing the second-highest SR on Datasets 4 and 5, and the third-highest on Dataset 3. We assume that this is due to its two-stage matching strategy and robust primary direction estimation based on norm feature.

Deep learning-based methods, SuperGlue and LoFTR, generally performed moderately on Datasets 2–5, yet still surpassed SIFT, OS-SIFT, RIFT, and CoFSM. ReDFeat, a network specifically designed for multimodal feature matching and trained on the Optical-SAR images of Dataset 2, achieved an RMSE slightly superior to MSG on this dataset. Its performance on Dataset 3 was second to MSG; however, a notable

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TABLE V

QUANTITATIVE COMPARISON OF EXPERIMENTAL RESULTS FOR THE RS AND CV TYPES ON DATASET 1 (BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST PERFORMANCE AMONG ALL METHODS, RESPECTIVELY)

Dataset type	Metric	Tested method										
		SIFT	OS-SIFT	RIFT	CoFSM	HOWP	POS-GIFT	SuperGlue	LoFTR	ReDFeat	MSG (ours)	MSG* (ours)
Optical-DSM	$r_s(\%)$	30.0	50.0	40.0	70.0	**100.0**	**100.0**	**100.0**	90.0	**100.0**	**100.0**	**100.0**
	N_{cm}	3	19	16	44	116	849	151	19	89	558	**999**
	e_{rms}	4.83	4.33	4.35	3.90	2.17	2.43	**1.51**	2.39	1.65	1.67	1.99
Optical-Infrared	$r_s(\%)$	30.0	80.0	70.0	60.0	**100.0**	**100.0**	90.0	80.0	**100.0**	**100.0**	**100.0**
	N_{cm}	7	71	59	361	189	1109	80	220	174	993	**1645**
	e_{rms}	5.22	3.19	3.13	3.02	1.57	1.77	2.19	2.41	1.49	**1.22**	1.67
Optical-Map	$r_s(\%)$	90.0	80.0	80.0	90.0	**100.0**	**100.0**	**100.0**	80.0	**100.0**	**100.0**	**100.0**
	N_{cm}	36	32	83	230	204	**1698**	174	111	183	911	1340
	e_{rms}	2.16	3.51	3.21	2.50	1.43	1.51	1.23	2.75	1.55	**1.10**	1.66
Optical-SAR	$r_s(\%)$	9.1	63.6	54.5	72.7	**100**	90.9	**100.0**	90.9	90.9	**100.0**	90.9
	N_{cm}	1	19	63	123	102	1073	85	190	100	884	**1464**
	e_{rms}	5.61	4.59	4.47	4.23	2.72	3.09	2.50	3.38	2.84	**2.38**	3.48
Day-Night	$r_s(\%)$	54.5	27.3	36.4	72.7	81.8	90.9	**100.0**	72.7	72.7	**100.0**	**100.0**
	N_{cm}	10	17	30	138	72	853	77	361	21	925	**1546**
	e_{rms}	4.62	5.16	4.77	4.42	3.20	3.03	2.42	3.68	3.55	**2.17**	3.08
Optical-Depth	$r_s(\%)$	14.3	42.9	**100.0**	71.4	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
	N_{cm}	2	16	131	264	120	**1702**	303	331	271	955	1562
	e_{rms}	5.50	4.92	1.87	3.34	2.31	1.78	**1.39**	2.55	1.43	1.65	2.35
Multi-temporal Optical	$r_s(\%)$	28.6	57.1	85.7	85.7	**100**	85.7	**100**	85.7	85.7	**100.0**	**100.0**
	N_{cm}	14	27	222	504	124	2067	200	866	205	1234	**2080**
	e_{rms}	4.79	3.69	2.59	2.37	2.49	1.85	2.06	2.03	2.36	**1.49**	2.11
RGB-NIR	$r_s(\%)$	96.7	**100.0**	86.7	70.0	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
	N_{cm}	388	687	660	279	730	2067	307	**4511**	493	1983	1892
	e_{rms}	1.82	1.77	2.33	4.04	1.85	**1.05**	1.67	1.64	1.90	1.64	1.86
VIS-IR	$r_s(\%)$	9.1	27.7	45.5	36.4	90.9	72.7	**100.0**	72.7	81.8	**100.0**	90.9
	N_{cm}	1	60	58	29	62	565	48	56	81	**876**	741
	e_{rms}	5.72	3.80	4.94	4.98	4.06	4.67	2.89	3.69	3.01	**2.44**	3.31

decline in performance was observed when tested on out-of-domain data. On Datasets 2–4, MSG's NCM was more than double that of all other methods. Furthermore, its SR values on Datasets 3–5 were 22.5%, 36.0%, and 16.5% higher, respectively, than the next best performing method. Additionally, the MSG* variant, achieved by minor modifications to the edge detection process and an increase in the descriptor radius, significantly enhanced the SR on Datasets 2 and 3.

As shown in Table V, Dataset 1 RGB-NIR is the easiest type to match. SIFT showed high SR on Optical-Map and RGB-NIR pairs, but had the lowest NCM and highest RMSE across other datasets due to its design not accounting for multimodal scenarios. Although OS-SIFT showed higher average SR and NCM than SIFT, the improvement was limited, and its performance was poorer in some datasets because it primarily targets SAR images. RIFT successfully matched most image pairs but struggled with pairs having significant scale differences, such as Optical-DSM, Optical-SAR, Day-Night, and RGB-NIR pairs, resulting in higher average RMSE. CoFSM displayed relatively balanced performance in the RS types but was less effective in the CV types, indicating weaker robustness. LoFTR showed high SR, but NCM varied widely, with some images having either too many or too few points concentrated. SuperPoint + SuperGlue exhibited the lowest RMSE in Optical-DSM and Optical-Depth types and generally performed well across most images. Compared to the Transformer-based LoFTR, it showed better cross-modal generalization, which we attribute to the large-scale nature of RS images where a single pixel often contains information from several meters of ground targets. Consequently, a global attention mechanism may be less effective than a localized CNN. However, it failed on images with significant modal differences, particularly those with large rotational angles. ReDFeat demonstrated performance intermediate between SuperGlue and LoFTR. HOWP and POS-GIFT performed similarly, matching most image pair. However, RMSE exceeded 2 pixels in most cases, and HOWP struggled with large rotation differences. MSG clearly outperformed other methods, being the only one to correctly match all nine types of image pairs. It achieved the lowest RMSE in six types and the second lowest in 1 type, the third lowest in the remaining 2. Also, MSG had the highest NCM in six types, 2 the second, and 1 the third. Although MSG* increased NCM, this was accompanied by an increase in RMSE, potentially due to the less precise localization of these additional matches.

The effective utilization of edge information at each step enables MSG to extract significant information from true corresponding points between the two images, even under severe NRD conditions. The well-designed MSG detector and descriptors showed robustness to NRD and SGD, and the two-stage matching strategy improved the NCMs and final matching accuracy. Therefore, MSG is highly effective for various multimodal image matching applications, particularly in MRSIM.

Table VI presents the MT comparison of different feature-based algorithms on the RS dataset. SIFT, implemented in

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TABLE VI
RUNTIME COMPARISON ON DATASET 1 RS TYPES

Metric	SIFT	OS-SIFT	RIFT	CoFSM	HOWP	POS-GIFT	MSG (ours)
$t_{\mathrm{m}}(s)$	1.31	11.84	23.81	19.37	6.33	30.53	9.01

C++, had the shortest MT but performed poorly in other quantitative metrics, making its low runtime less meaningful. HOWP had a shorter runtime but significantly dropped in performance or failed with large rotational differences. OS-SIFT and CoFSM consumed more time during scale space construction and feature point extraction and had much lower NCM compared to MSG. RIFT was the second most time-consuming due to its ring structure, which simulates image rotation for rotation invariance. POS-GIFT was the most time-consuming among all methods, using PC to construct descriptors and reconstructing descriptors in the POS stage, which increased NCM but also computation time. MSG had a relatively short runtime, slower only than SIFT and HOWP. We used separable filters for scale-space construction and parallel computing for descriptor construction. Most time was spent in mismatch removal due to the higher NCM.

2) Qualitative Evaluation: We selected representative image pairs from four datasets for qualitative evaluation. These image pairs present significant modal and geometric differences, making precise matching challenging. To further validate the performance of the proposed method, this section visualizes the matching and registration results of MSG.

Figs. 9 and 10 display the matching results of MSG across RS types of Dataset 1 and Datasets 2–5, with five images selected from each type or dataset. MSG produced a large number of correctly matched points evenly distributed across the entire image, effectively suppressing noise interference, SAR, and NRD in multimodal images, achieving scale and rotation invariance. This demonstrates the significant application potential of the MSG algorithm in MRSIM tasks.

Figs. 11 and 12 compare the matching results of MSG with nine other algorithms. On Dataset 1 (Fig. 11), SIFT showed the poorest matching performance. Although OS-SIFT found some correct matches in images with significant modal differences, it ultimately failed in those cases. RIFT, based on PC, extracted features from maximum and minimum moment images, balancing the number and repeatability of features. However, its lack of scale invariance led to decreased performance or failure with even small scale differences. CoFSM produced many matches in some images but failed completely in others. HOWP showed excellent performance with small rotation differences but failed with large rotation differences and significant modal differences. Learning-based methods also failed when image pairs have large rotation differences or complex NRD. POS-GIFT used the POS inlier recovery strategy for descriptor reconstruction and secondary matching, achieving the highest NCM in most images but with too concentrated feature points, increasing RMSE. It also showed instability with the large modal differences in Optical-SAR and Day-Night pairs. MSG achieved robust matching in images with geometric distortions and NRD, with both scale and rotation invariance, and relatively low time consumption. These results further validate the effectiveness and robustness of MSG. On Datasets 2–5 (Fig. 12), all methods had low SR for all effects except MSG.

TABLE VII
PERFORMANCE OF MSG ON DATASET 1 RS TYPES WITH VARYING NUMBERS OF OCTAVES AND LAYERS IN MSGSS

Method	$r_{\mathrm{s}}(\%)$	N_{cm}	e_{rms}	$t_{\mathrm{m}}(s)$
MSGSS (1 octave with 4 layers)	100	907	1.69	9.01
1) 1 octave with 5 layers	100	1063	1.77	11.50
2) 2 octaves with 4 layers	100	899	1.75	11.72
3) 3 octaves with 4 layers	100	1213	1.76	12.83

Fig. 13 shows the registration results of the MSG algorithm using a checkerboard visualization. The overlapping regions between the images are well-aligned in the checkerboard pattern. The visual results are consistent with the quantitative calculations based on real corresponding points, achieving subpixel registration accuracy within 2 pixels, demonstrating the high precision of matching points in MRSIM tasks.

D. Rotation and Scale Invariance

1) Scale Invariancce: To evaluate the scale invariance of MSG, a set of Optical-Map and Optical-SAR image pairs was tested. The original images were resampled to generate six sets of simulated images with different scales (0.25×, 0.33×, 0.5×, 2×, 4×, and 6×). The matching results, shown in Fig. 14, demonstrate that as the scale difference increased, a large number of evenly distributed correct matched points were still detected. Since the descriptor radius only differs by a factor of 2 across octaves, the four layers can handle a scale difference up to 2×. Further scale differences can be addressed by adding more layers within an octave or increasing the number of octaves in the scale pyramid through image downsampling. The results in Fig. 14 use a three-octave pyramid (with downsampling factors of 1, 1/2, and 1/4). Table VII compares the performance of MSG with different scale spaces. Expanding the scale space resulted in a runtime increase of over 30% and a slight rise in RMSE. A scale-space configuration with one octave and four layers ensures high matching accuracy while maintaining optimal computational efficiency, which we use as the default setting.

2) Rotation Invariance: To verify the rotation invariance of MSG, five image pairs from Dataset 1 were selected for demonstration and the parameter R_s is set to 20. One image in these image pairs was rotated counterclockwise at 30° intervals to generate 11 directions from 30° to 330°. The matching results are shown in Fig. 15. MSG successfully matched the images in all directions, with rich and evenly distributed correct matching points, showing little variation in the NCM.

E. Ablation Studies

1) Filter of Scale Space: To validate the effectiveness of the proposed MSGSS, we conducted experiments on Dataset

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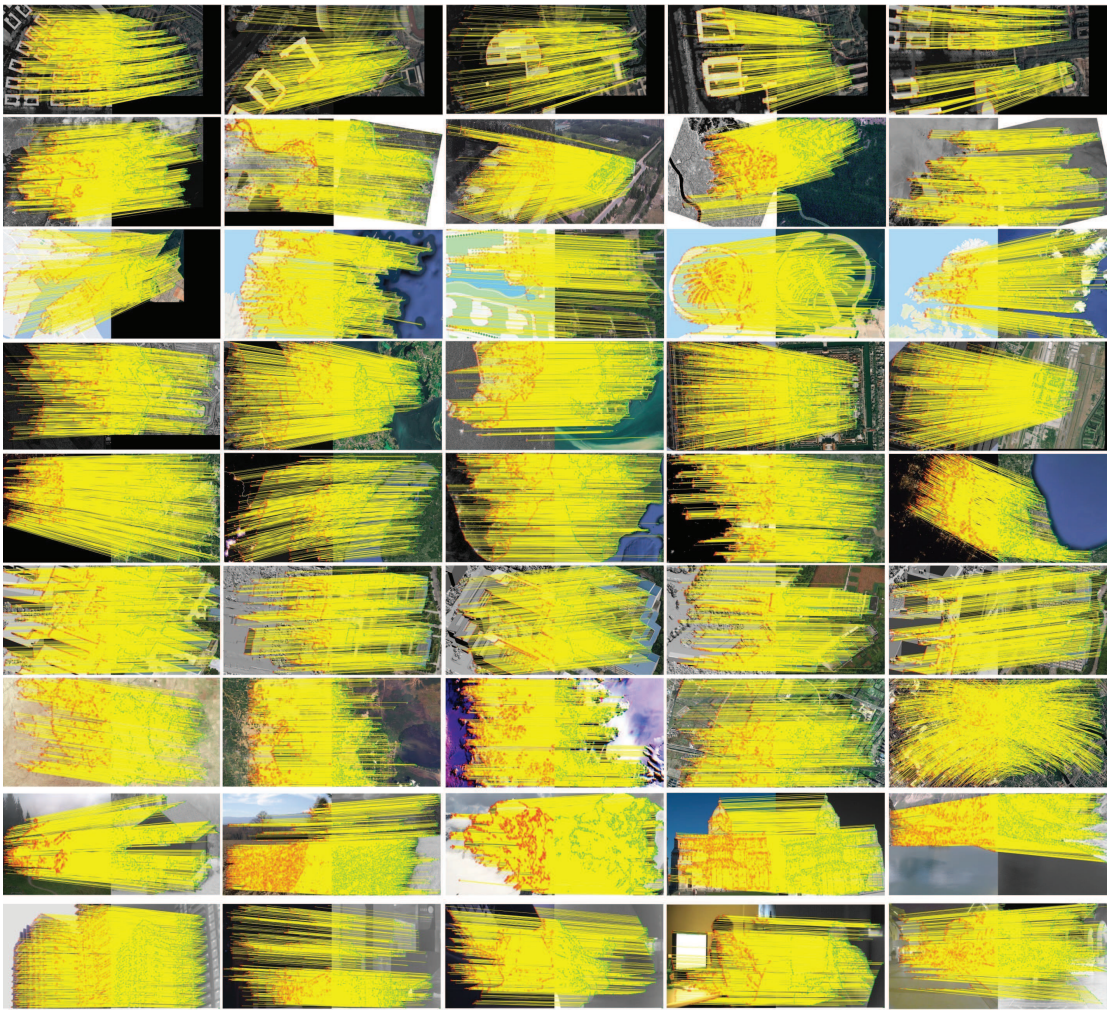

Fig. 9. Matching results of MSG on nine types of multimodal image pairs on Dataset 1. The image types in order from the first to the last row are Optical-DSM, Optical-Infrared, Optical-Map, Optical-SAR, Day-Night, Optical-Depth, Multitemporal Optical, RGB-NIR, and VIS-IR.

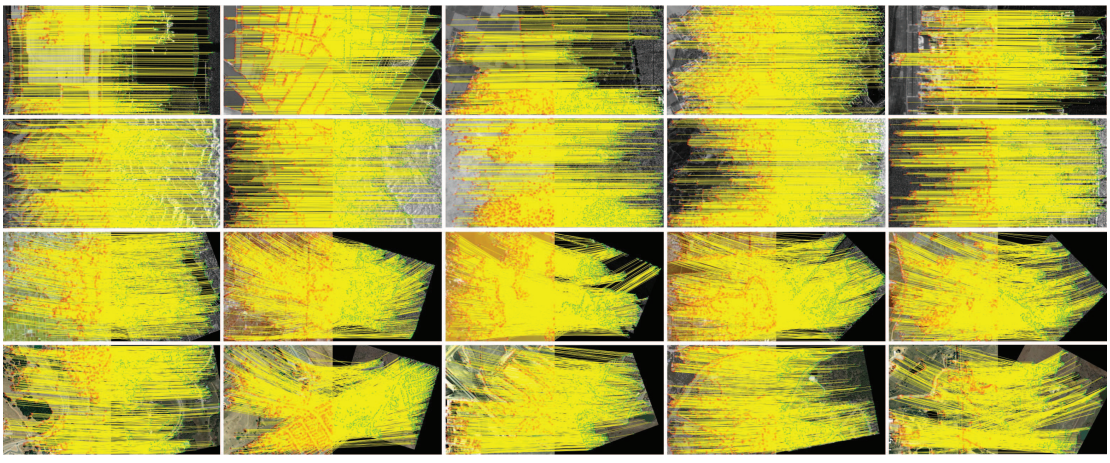

Fig. 10. Matching results of MSG on Dataset 2–5. Each row shows one dataset in order.

1 RS types. We replaced MSGSS with constructions from co-occurrence filtering (CoF) [15], local normalization filtering (LNF) [34], and pointwise shape-adaptive texture filtering (PSATF) [60] while keeping other parts unchanged. The parameters used for these methods were taken from the respective papers. Additionally, we replaced the filters in MSGSS with GF and reduced the number of S-GAU's orientations to 8 for comparison. The results are shown in Table VIII.

MSGSS retains edge information across different scales to the greatest extent, resulting in the highest matching accuracy

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Fig. 11. Matching results of different methods on Dataset 1 RS types are shown, and the connecting line is red for matching failure, from top to bottom: SIFT, OS-SIFT, RIFT, CoFSM, HOWP, POS-GIFT, SuperPoint + SuperGlue, LoFTR, ReDFeat, and MSG (ours).

and precision among all compared methods. Furthermore, for an $m \times n$ sized Gaussian filter, the separability of the filter significantly accelerates computational efficiency, with an acceleration ratio of $(m \times n/m + n)$.

When replaced with GF, the matching performance significantly declined, with a 16.7% match failure rate and a reduction in the NCM to half of MSGSS. This result underscores the critical role of MSGSS in enhancing edge information and effectively capturing multiscale features.

LNF, which uses the input image subtracted by the result of mean filtering, reduces the differences between images. However, it significantly decreases the usable image features and has poor resistance to noise, leading to the worst performance among the methods tested.

Both the CoF and PASTF are based on the statistical properties of the image. They preserve the image structure while removing internal texture details, but their computational complexity is higher. The CoF computes the co-occurrence matrix, which is the most time-consuming process. While CoF yields relatively good results, it blurs the edges more than MSGSS, producing halo artifacts. PASTF results in excessive smoothing, and edge distortion occurs, which hampers the preservation of edge position and shape. As a result, PASTF performs worse than CoF. However, both methods are more effective than GF and LNF in maintaining edge information, which is crucial for MRSIM.

The S-GAU with eight orientations provides results close to MSGSS, but its matching performance deteriorates in images with significant scale differences. As the number of layers in the scale space increases, the rounds of filtering operations also increase, which leads to a reduction in the edge information in the S-GAU eight-orientation filtered images.

2) Keypoint Detection: In this section, a comparative test was conducted to evaluate the performance of different keypoint detectors. Assuming that the detector extracts M and N keypoints from the two images, C_{matched} is defined as the number of matching points extracted simultaneously in both images within a 3 pixel error under the ground truth

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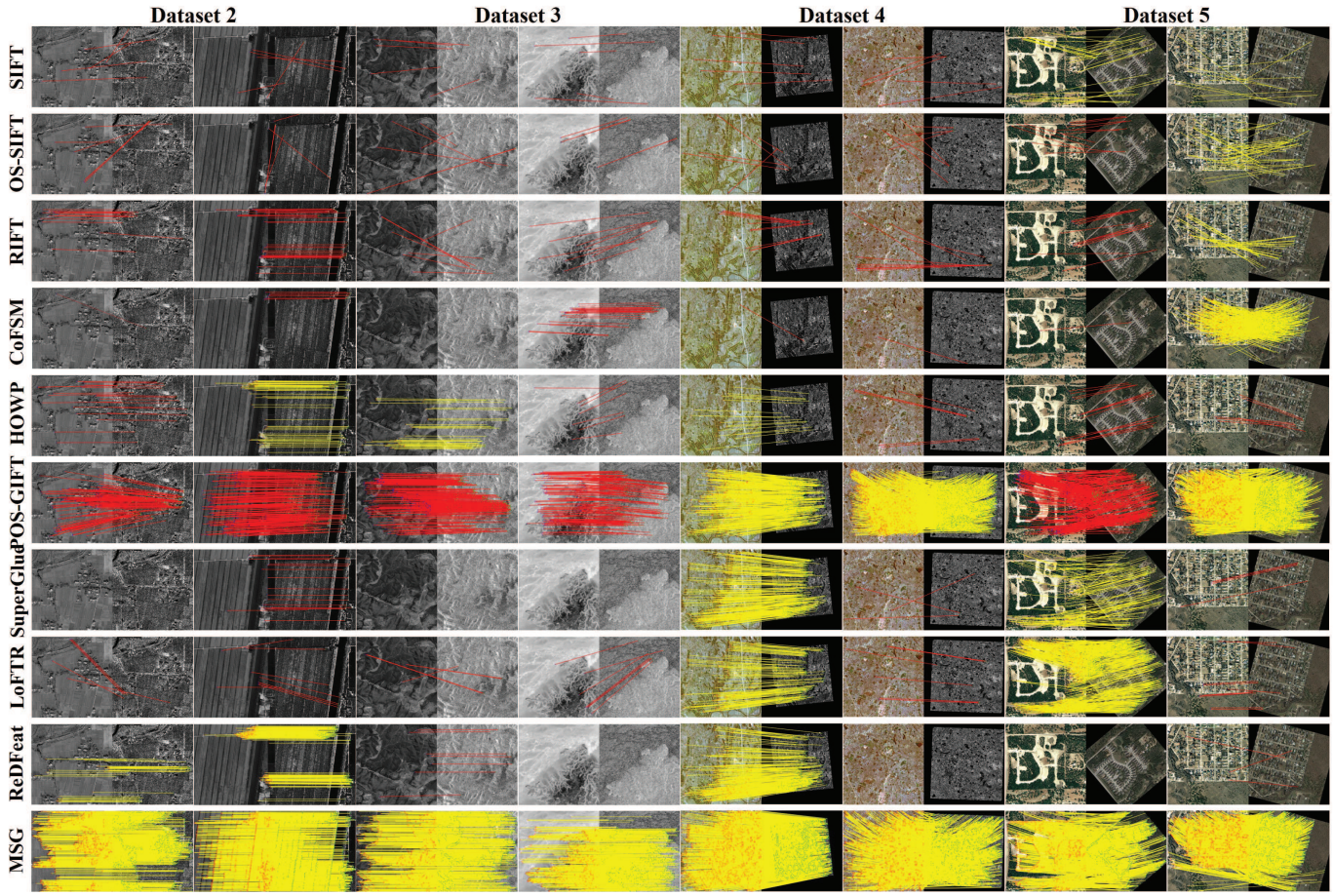

Fig. 12. Matching results of different methods on Dataset 2–5 are shown, and the connecting line is red for matching failure, from top to bottom: SIFT, OS-SIFT, RIFT, CoFSM, HOWP, POS-GIFT, SuperPoint + SuperGlue, LoFTR, ReDFeat, and MSG (ours).

transformation. The repeatability (R_{p}) is defined as

$$R_{\mathrm{p}} = \frac{C_{\mathrm{matched}}}{(M+N)/2}. \tag{19}$$

We compared four methods: Harris corner detection on the S-GAU scale space's edge map (SGAUE), Harris corner detection on the S-GAU scale space's image (SGAU), Harris corner detection on the Gaussian scale-space image (GAU), and FAST corner detection on the maximum and minimum moment images obtained from the PC map [22], selecting the top 5000 points (PC-FAST) [13]. As shown in Fig. 16, on the seven sets of RS images, GAU had the lowest repeatability in six types of images, SGAU outperformed GAU overall and was comparable to PC-FAST, and SGAUE achieved the best performance across all seven types of RS images. This is because MSGSS enhances edge information, and MSG combines keypoints from different scales. Therefore, SGAUE is capable of obtaining highly repeatable keypoints, which is crucial for ensuring accurate matching points in the subsequent matching steps.

3) Others: To verify the effectiveness of other components of MSG, ablation experiments were conducted on Dataset 1 RS types by replacing different components while keeping others constant. The specific results are shown in Table IX.

1) When the SIFT-like gradient feature was used, the ability to capture gradient variations at the edges of different modality images was reduced, resulting in performance degradation.
2) Using the second-order Gaussian steerable filter [30] for gradient feature extraction did not improve the performance. On the contrary, it blurred the phase information of the image because of its higher order information, causing a significant drop in gradient performance.
3) A single-stage mismatch removal process achieved over 90% SR and resulted in fewer NCM. The two-stage mismatch removal process integrated into MSG improved both the accuracy and robustness of the matching results.
4) Furthermore, relying solely on descriptor distance for matching reduced the NCM and slightly increased RMSE. The combined distance metric employed in MSG proved more effective, enhancing matching accuracy.
5) Using the Canny operator for edge detection increased NCM but caused failures in some Optical-Map image pairs due to sparse texture. In these cases, the concentration of keypoints in localized areas of the map data, coupled with the distribution of keypoints across the entire optical image, led to many initial mismatches.

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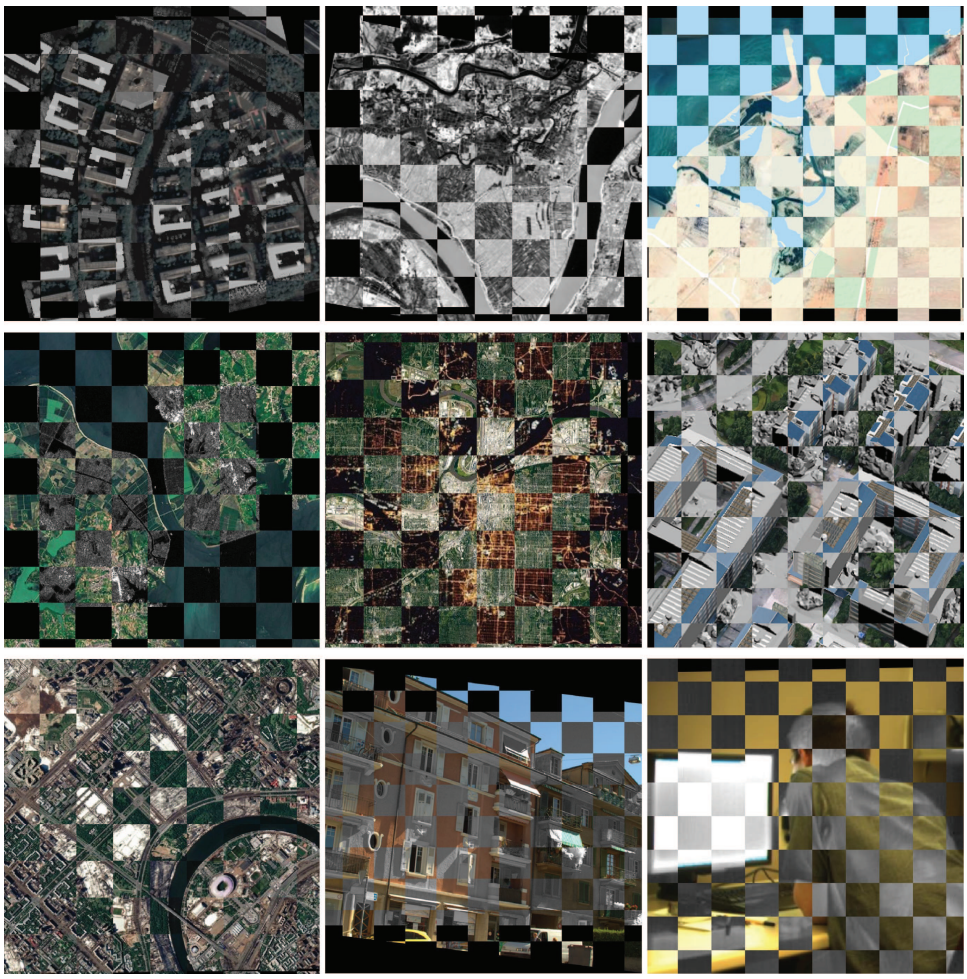

Fig. 13. Partial registration results of MSG.

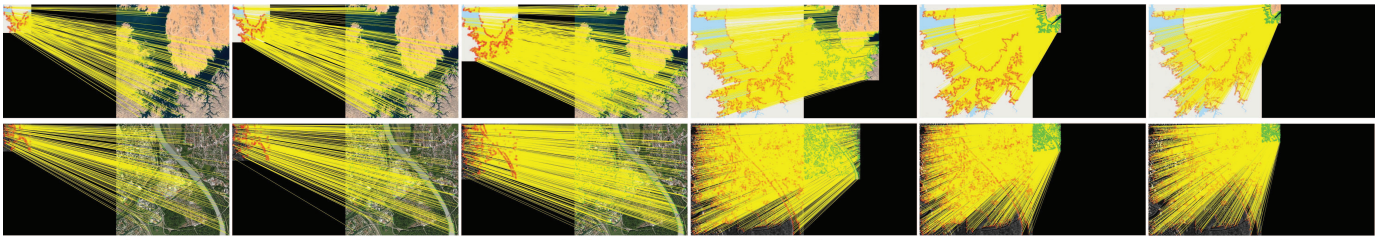

Fig. 14. Scale invariance demonstration.

Increasing the descriptor radius could help address this issue.

F. Applications in Geometric Registration

To further validate the matching performance and generalization capability of our MSG on more practical datasets, we performed geometric registration on multiscene, large-scale, multimodal images and conducted a qualitative analysis of MSG's applicability.

The selected experimental datasets are intended to comprehensively evaluate the algorithm's performance under various complex scenarios. A summary description of the datasets is provided in Table X and Fig. 17. The first group is sourced from [30], while groups 2–4 are collected by the researchers and include optical and SAR data from different regions and scenes. These datasets exhibit significant differences in spatial resolution, NRD, and severe geometric distortions (including layover, foreshortening, and radar shadow effects specific to SAR imagery) between different modalities, and variations in original acquisition dates. These challenging characteristics fully demonstrate the adaptability of MSG to complex real-world scenarios.

The algorithm parameters were set identically to those for Datasets 2–5 in Section IV-C. We divided the overlapping regions of the input images into multiple image patches of

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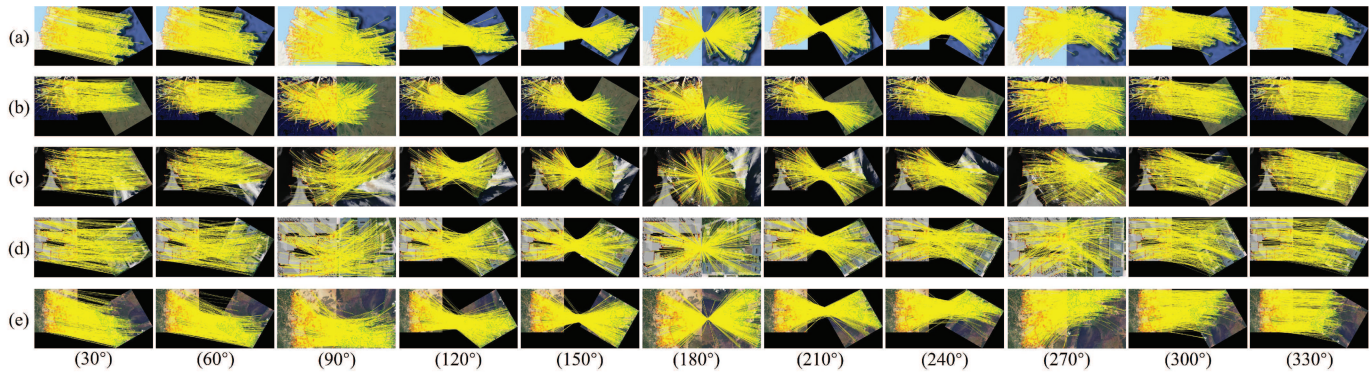

Fig. 15. Rotation invariance demonstration. (a) Optical-Map. (b) Optical-SAR. (c) Day-Night. (d) Optical-Depth. (e) Multitemporal Optical.

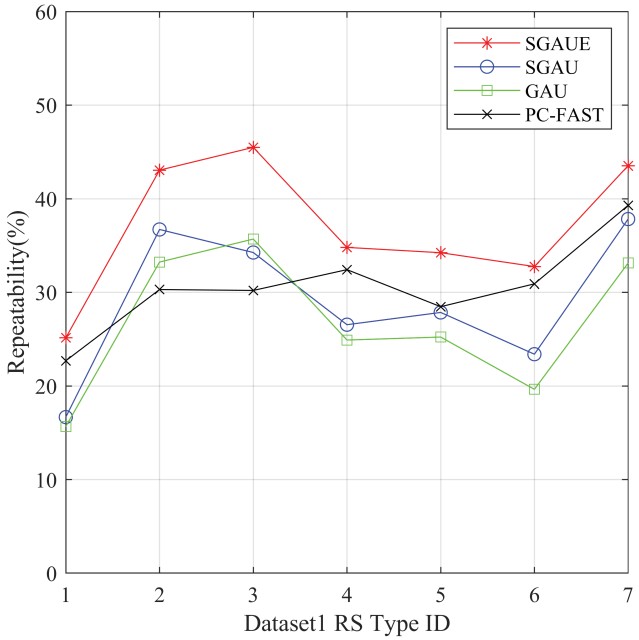

Fig. 16. Comparison of repeatability of different keypoint detection methods on Dataset 1 RS types.

TABLE VIII
RESULTS OF ABLATION EXPERIMENTS ON FILTER TYPES IN SCALE SPACE FOR DATASET 1 RS TYPES

Method	$r_s(\%)$	N_{cm}	e_{rms}	$t_m(s)$
MSGSS	100	907	1.69	9.01 (0.44)
1) Gaussian space	83.3	449	2.56	4.98 (0.06)
2) LNF space	78.8	211	2.98	7.99 (0.04)
3) CoF space	98.5	591	1.95	17.5 (11.04)
4) PASTF space	92.4	888	2.32	13.4 (5.95)
5) S-GAU with 8 orientation	98.5	899	1.81	8.74 (0.31)

The values in parentheses in the column t_m(s) represent the computational time for constructing scale space.

2048 × 2048 pixels, and each patch was further downscaled to 1024 × 1024 pixels. After local matching was completed for all image patches, rectification was implemented using Python. Considering that successful matching cannot be guaranteed in all regions, we employed the RANSAC algorithm with the *estimateAffine2D* function to remove outliers from all matching points, setting the threshold to 20 pixels. Finally, from the remaining matching points, we selected control points by sampling every 20th point. Image rectification was performed using the second-order polynomial model in the *gdal.Warp* function.

TABLE IX
RESULTS OF ABLATION EXPERIMENTS ON OTHER COMPONENTS FOR DATASET 1 RS TYPES

Method	$r_s(\%)$	N_{cm}	e_{rms}	$t_m(s)$
Full MSG	100	907	1.69	9.01
1) Remove first-order Gaussian steerable filtering	95.5	873	1.98	9.40
2) Replace first-order steerable filtering with second-order	71.2	756	3.10	9.11
3) Replace two-stage match strategy with FSC	90.9	149	2.46	5.05
4) Replace new distance with Euler distance	100	766	1.76	8.83
5) Replace Sobel with Canny operator in edge extraction	97.0	1594	1.93	13.61

Fig. 18 illustrates the matching points detected by MSG in all test cases, along with the distribution of their error magnitudes and directions. It can be clearly observed that despite significant NRD, degree of geometric distortion, noise levels, and spatial resolution among each multimodal image pair, our MSG is still able to detect a sufficient number of well-distributed matching points in most pairs. This demonstrates the algorithm's strong robustness against various complex challenges. Meanwhile, the overall error distribution trend is consistent with visual inspection results, indicating high matching accuracy.

To intuitively examine the final registration results, Fig. 19 presents the overlay results of the reference and sensed images before and after registration. Evidently, all sensed images are correctly aligned with the reference images, with clear feature edges and well-matched geometric structures. However, it should be noted that in certain specific regions—such as areas with severe SAR layover and radar shadow in mountainous regions (leading to information loss or severe geometric distortion), large water body areas (which typically exhibit weak texture, and lack discernible features), and large weak

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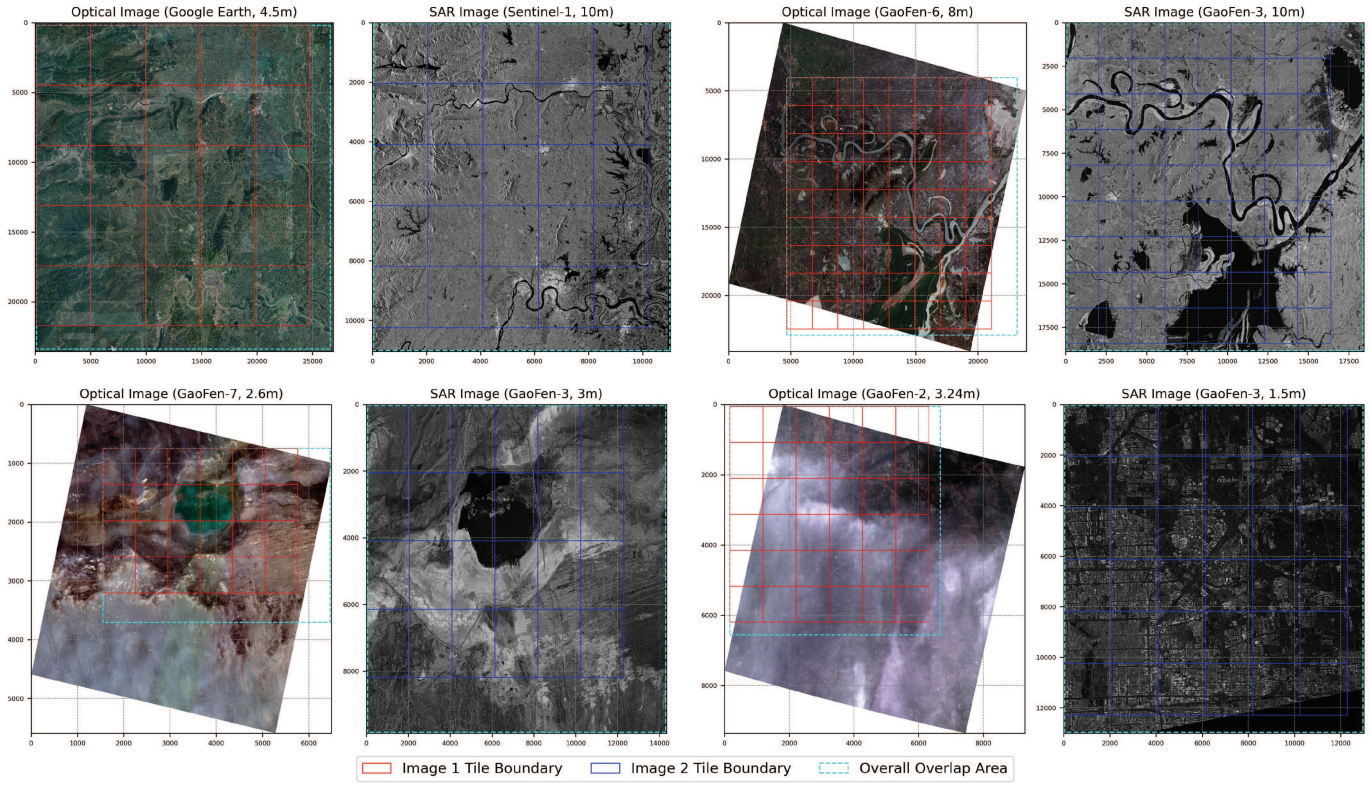
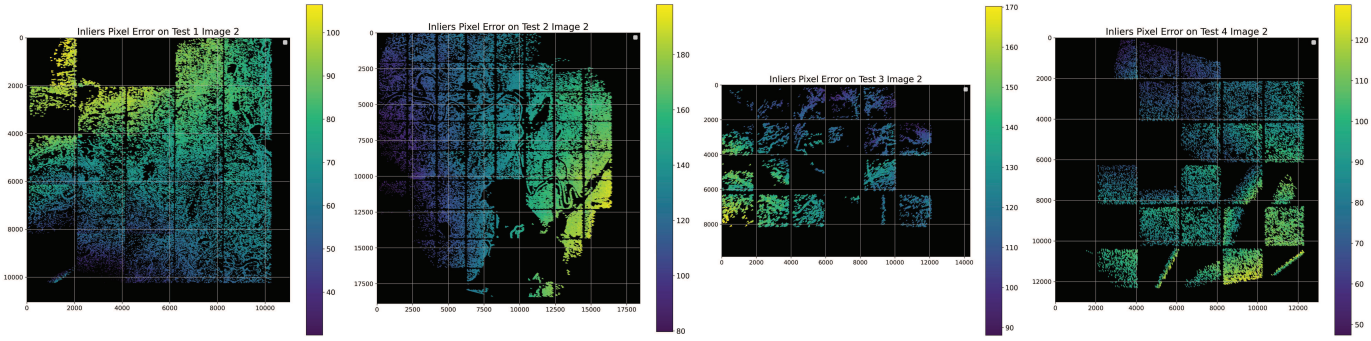


Fig. 17. Test image pairs for geometric registration, also including the image patching strategy employed in the experiments.



(a) (b) (c) (d)

Fig. 18. Test image pairs, with error magnitude and direction distribution for image 2. Arrow color and size depict the MSG's pixel matching error magnitude, and arrow orientation indicates the error direction (zoom in for clearer view of arrow directions). (a) Test 1. (b) Test 2. (c) Test 3. (d) Test 4.

texture desert areas in Dunhuang city—the matching errors are relatively larger. These regions pose significant challenges to registration due to inherent feature sparsity, texture ambiguity, or severe geometric distortions. The matching SR in these areas is relatively low, necessitating parameter adjustments to achieve better results (e.g., increasing the number of points and descriptor radius while scaling the image to encompass more texture). Nevertheless, the proposed MSG can still provide acceptable registration results in these difficult areas, laying a foundation for subsequent higher precision processing.

G. Limitations and Prospects

The proposed MSG demonstrates commendable performance by leveraging robust edge information, achieving acceptable registration results across various scenarios as detailed previously. However, its efficacy can be constrained in regions with extensive weak textures, such as deserts or large water bodies, where the scarcity of discernible features limits initial match generation and descriptor distinctiveness. Similarly, areas with highly repetitive patterns, like agricultural farmlands, or where there are inconsistencies in local edges, like large-scale land cover changes, may lead to ambiguous correspondences. While parameter optimization can alleviate some of these issues, these represent inherent challenges for feature-based approaches.

Future research will focus on enhancing practical applicability through refined parameter tuning for specific project deployments and utilizing the algorithm for assisted registration workflows. Key prospects include the development of large-scale, diverse multimodal RS datasets (encompassing

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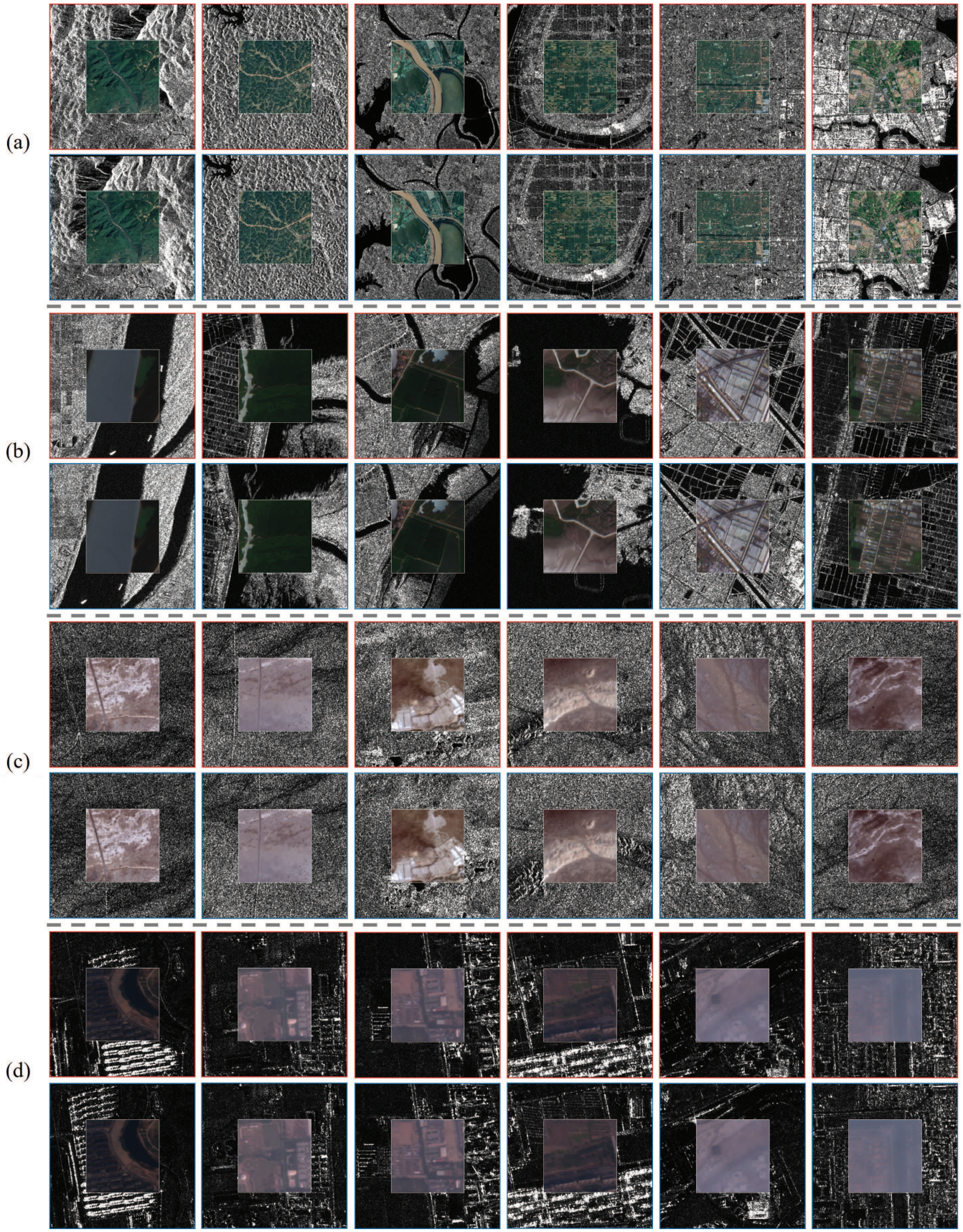

Fig. 19. Comparative visualization of image patch overlays before (the upper panel) and after (the lower panel) registration. Each subfigure displays the optical image patch centrally, with the corresponding SAR image patch peripherally overlaid. (a) Test 1. (b) Test 2. (c) Test 3. (d) Test 4.

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TABLE X
SUMMARY DESCRIPTIONS OF THE TEST IMAGES

Test case	Reference image	Sensed image	Image characteristic
Test1	Sensor: Google Earth, Optical Spatial resolution: 4.5 m Data: 08/2016 Size: 26880 × 23552	Sensor: Sentinel-1, SAR Spatial resolution: 10 m Data: 09/2018 Size: 10980 × 10980	This dataset is located in Hubei, China. The west of images is covered by mountains, while the east covers a suburban area with rivers and buildings. Moreover, the image resolution difference is about two times.
Test2	Sensor: GaoFen-6, Optical Spatial resolution: 8 m Data: 04/2025 Size:23893 × 24096	Sensor: GaoFen-3, SAR Spatial resolution: 10 m Data: 08/2024 Size: 18431 × 18877	This dataset is situated in the Jianghan Plain, China, characterized by large-scale regular farmlands and a dense network of rivers and lakes. The periodic patterns of farmlands create repetitive textures, while rivers and lakes fall under weak-texture water bodies.
Test3	Sensor: GeoFen-7, Optical Spatial resolution: 2.6 m Data: 05/2025 Size: 6489 × 5589	Sensor: GaoFen-3, SAR Spatial resolution: 3 m Data: 06/2023 Size: 14334 × 9860	This dataset is located in Dunhuang, China, a typical arid/semi-arid region, encompassing vast desert areas and little oasis landforms, where sand dunes are typical weak-texture regions.
Test4	Sensor: GeoFen-2, Optical Spatial resolution: 3.24 m Data: 04/2025 Size: 9274 × 9355	Sensor: GaoFen-3, SAR Spatial resolution: 1.5 m Data: 02/2025 Size: 13000 × 13000	This dataset centers on a typical urban core in Beijing, China, containing numerous high-rise buildings and dense road networks. The images exhibit potential SAR layover caused by buildings and disparities arising from the differing acquisition perspectives.

optical, SAR, thermal infrared, glimmer, and map data), potentially augmented by image generation models to create controlled, simulated images with reduced noise. Furthermore, the integration of semantic understanding from advanced deep learning models, especially vision foundation models, and the design of more sophisticated outlier rejection strategies are anticipated to significantly improve robustness in these challenging contexts.

V. CONCLUSION

This article addresses the challenges posed by strong NRD and SGD in MRSIM by introducing a novel matching method, MSG, which leverages salient edge information to enhance matching performance. MSG enhances edge details by constructing a scale space based on side window GF and detects uniformly distributed, highly repeatable keypoints on edge maps using Harris corner detection and SSC. The algorithm improves rotation invariance by combining steerable filtering with the Sobel operator to obtain second-order gradients and achieves scale invariance by constructing improved GLOH descriptors of varying radii. The two-stage matching strategy with a constrained search space further improves the robustness and accuracy of the matching process. Experiments demonstrate that MSG outperforms nine other matching algorithms (SIFT, OS-SIFT, RIFT, CoFSM, HOWP, POS-GIFT, SuperPoint + SuperGlue, LoFTR, and ReDFeat) on five datasets. Notably, MSG consistently achieves higher NCM, particularly for RS type, and maintains the lowest RMSE. Its performance in handling scale and rotation differences has been further validated. MSG has also applied satisfactory practical registration on multimodal datasets collected in various regions and scenarios. The flexibility of MSG's parameters and matching strategies allows it to be adjusted for different task requirements.

However, MSG has limitations. Its reliance on edge and texture features may lead to suboptimal performance in regions with weak textures or large-sacle land cover changes. Additionally, the method faces challenges in matching images with significant projective distortion (e.g., satellite images versus UAV images). Future work will focus on developing more robust descriptors capable of handling weaker textures and greater modal differences in RS images, as well as improving the algorithm's computational efficiency. Additionally, the principles underlying MSG could be applied to guide the development of deep learning frameworks. Future efforts will also aim to create new MRSIM datasets to support training and testing in deep learning applications, addressing issues such as projection problems caused by large viewpoint changes.

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